



Mohamed Sathak
A J College of Engineering

An Autonomous Institution

SATHAK - CHENNAI



Approved by AICTE and UGC (12B & 2f)
Affiliated to Anna University
Accredited by NAAC Grade A & NBA
An ISO:21001-2018 Certified Institution

INDUSTRIAL SAFETY

II YEAR

(COMMON TO ALL)

NOTES

STUDENT NAME :

REG- NO :

DEPARTMENT :

INDUSTRIAL SAFETY			
Course Code		Course Type	Theory
Teaching Periods/Week (L: T:P)	3:0:0	Credits	
Total Teaching Periods	45 Periods	IAT + ESE Marks	
Teaching Department	Mechanical Engineering		
Course Objectives:			
• To study the fundamental concept and principles of industrial safety • To study the principles of maintenance engineering. • To Analyzing the wear and its reduction. • To study the faults in various tools, equipments and machines. • To study the periodic maintenance procedures in preventive maintenance.			
Unit: I	INDUSTRIAL SAFETY		9 Periods
Accident, causes, types, results and control, mechanical and electrical hazards, types, causes and preventive steps/procedure, describe salient points of factories act 1948 for health and safety, wash rooms, drinking water layouts, light, cleanliness, fire, guarding, pressure vessels, etc, Safety color codes. Fire prevention and firefighting, equipment and methods.			
Teaching-Learning Process		Pedagogy: Lecture, PPT	
		RBT Level: L1, L2, L3	
Unit: II	MAINTENANCE ENGINEERING		9 Periods
Definition and aim of maintenance engineering, Primary and secondary functions and responsibility of maintenance department, Types of maintenance, Types and applications of tools used for maintenance, Maintenance cost & its relation with replacement economy, Service life of equipment.			
Teaching-Learning Process		Pedagogy: Lecture, PPT	
		RBT Level: L1, L2, L3	
Unit: III	WEAR AND CORROSION AND THEIR PREVENTION		9 Periods
Wear- types, causes, effects, wear reduction methods, lubricants-types and applications, Lubrication methods, general sketch, working and applications, i. Screw down grease cup, ii. Pressure grease gun, iii. Splash lubrication, iv. Gravity lubrication, v. Wick feed lubrication vi. Side feed lubrication, vii. Ring lubrication, Definition, principle and factors affecting the corrosion. Types of corrosion, corrosion prevention methods.			
Teaching-Learning Process		Pedagogy: Lecture, PPT	
		RBT Level: L1, L2, L3	
Unit: IV	FAULT TRACING		9 Periods
Fault tracing-concept and importance, decision tree concept, need and applications, sequence of fault finding activities, show as decision tree, draw decision tree for problems in machine tools, hydraulic, pneumatic, automotive, thermal and electrical equipment's like, i. Any one machine tool, ii. Pump iii. Air compressor, iv. Internal combustion engine, v. Boiler, vi. Electrical motors, Types of faults in machine tools and their general causes.			
Teaching-Learning Process		Pedagogy: Lecture, PPT	
		RBT Level: L1, L2, L3	
Unit: V	PERIODIC AND PREVENTIVE MAINTENANCE		9

													Periods		
Periodic inspection-concept and need, degreasing, cleaning and repairing schemes, overhauling of mechanical components, overhauling of electrical motor, common troubles and remedies of electric motor, repair complexities and its use, definition, need, steps and advantages of preventive maintenance. Steps/procedure for periodic and preventive maintenance of: i. Machine tools, ii. Pumps, iii. Air compressors, iv. Diesel generating (DG) sets, Program and schedule of preventive maintenance of mechanical and electrical equipment, Advantages of preventive maintenance. Repair cycle concept and importance															
Teaching-Learning Process		Pedagogy: Lecture, PPT													
		RBT Level: L1, L2, L3													
Total													45 Periods		
Suggested Activities and Evaluation Methods: Case Study, Quiz, Assignment topics, Class Presentation and Continuous Assessment Test															
Course Outcomes:															
After successful completion of this course, the students should be able to															
CO1: Explain the fundamental concept and principles of industrial safety															
CO2: Apply the principles of maintenance engineering.															
CO3: Analyze the wear and its reduction.															
CO4: Evaluate faults in various tools, equipment's and machines															
CO5: Apply periodic maintenance procedures in preventive maintenance.															
Text Books:															
T1 L. M. Deshmukh, Industrial Safety Management, Tata McGraw-Hill Education, 1st Edition, 2005.															
T2 Charles D. Reese, Occupational Health and Safety Management: A Practical Approach, CRC Press, 3rd Edition, 2015.															
References															
R1 Edward Ghali, V. S. Sastri, M. Elboudjaini, Corrosion Prevention and Protection: Practical Solutions, John Wiley & Sons, 2007.															
R2 Garg, HP, Maintenance Engineering, S. Chand Publishing, 2012															
R3 J Maiti, Pradip Kumar Ray, Industrial Safety Management: 21st Century Perspectives of Asia, Springer, 2017															
R4 R. Keith Mobley, Maintenance Fundamentals, Elsevier, 2011.															
Web links and Video Lectures (e-Resources):															
1. https://www.youtube.com/watch?v=v-eltsixu4I															
2. https://www.youtube.com/watch?v=jFDWlKayrTc&list=PLbRMhDVUMngdXebaRB59KdKwstzuAovua															
3. https://www.youtube.com/watch?v=ZEShNJX3kcg&list=PLbRMhDVUMngdXebaRB59KdKwstzuAovua&index=12															
CO	PO												PSO		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
1	2	1	2			2	1					1	1	2	1
2	2	1	2			2	1					1	1	2	1
3	2	1	2			2	1					1	1	2	1
4	2	1	2			2	1					1	1	2	1
5	2	1	2			2	1					1	1	2	1
Low (1) ; Medium (2) ; High (3)															

Unit: I: INDUSTRIAL SAFETY

INDUSTRIAL ACCIDENT

- According to the Factories Act, 1948: “It is an occurrence in an industrial establishment causing bodily injury to a person that makes them unfit to resume their duties in the next 48 hours”.
- In other words, accident is an unexpected event in the course of employment which is neither anticipated nor designed to occur. Thus, an accident is an unplanned and uncontrolled event in which an action or reaction of an object, a substance, a person, or a radiation results in personal injury.
 - It is important to note that self-inflicted injuries cannot be regarded as accidents. Why do Accidents Occur?
 - Lack of risk awareness.
 - Lack of knowledge on the activity being undertaken.
 - Lack of safety aspects in design.
 - Lack of commitment to safety.
 - Lack of control.
 - Lack of education, training and motivation.
 - Lack of team-work and safety culture.
 - Lack of discipline.
 - Lack of social responsibility in general and personal responsibility and accountability to safety.
 - Failure to learn from past experiences of similar incidents.
 - Failure to inspect safety gadgets and devices and maintain them in order regularly and adequately.

1. State-of-the-Art Regulations but Poor Implementation:

- India has established numerous laws to ensure workplace safety, including the Factories Act, 1948, which mandates safe working conditions, and the Occupational Safety, Health and Working Conditions Code, 2020, which aims to consolidate and update previous labor laws.
- Despite these regulations, workplace accidents remain high. According to the National Crime Records Bureau (NCRB) data, thousands of industrial accidents occur annually, leading to numerous fatalities and injuries.

2. Indifference by Government, Management, and Workers' Unions:

- **Government:** Often criticized for inadequate inspection and enforcement. Reports by the

Comptroller and Auditor General (CAG) have highlighted lapses in workplace safety inspections.

- **Management:** Many companies prioritize profit over safety, leading to unsafe working conditions. For instance, the Bhopal Gas Tragedy in 1984 remains a stark reminder of corporate negligence in India.
- **Workers' Unions:** While they advocate for workers' rights, their influence on improving safety standards is limited. They often focus more on wage-related issues rather than safety.

3. **Stringent but Ineffective Statutory Provisions:**

- The legal framework is robust on paper, with severe penalties for non-compliance. For example, the Occupational Safety, Health and Working Conditions Code, 2020, includes fines and imprisonment for violations.
- However, these provisions are seldom enforced. A study by the International Labour Organization (ILO) found that many companies fail to comply with safety standards, and enforcement mechanisms are weak.

4. **Lack of Awareness Among Implementers:**

- Officials responsible for enforcing safety regulations often lack the necessary training and resources. The ILO report also noted a significant gap in the training of inspectors and lack of modern equipment for safety inspections.

5. **Bureaucratic Corruption:**

- Corruption within regulatory bodies exacerbates the issue. Transparency International's Corruption Perceptions Index often ranks India poorly, indicating widespread corruption.
- Instances of management bribing officials to overlook safety violations are reported frequently. This corruption undermines the enforcement of safety laws.

By addressing these issues—improving enforcement, reducing corruption, and increasing awareness and training among implementers—India can better translate its stringent safety regulations into effective workplace safety practices.

1. **Commitment to Safety by Management:**

- **Paper vs. Action:** Many companies have safety policies outlined in manuals, but these are often not implemented effectively. According to a 2022 report by the National Safety Council, nearly 60% of workers in the US felt their company's safety policies were not enforced consistently.
- **Core Culture:** Making safety a core part of the organizational culture involves regular

training, clear communication of safety protocols, and leadership that prioritizes safety. Research published in the Journal of Safety Research highlights that companies with strong safety cultures have significantly lower injury rates.

- **Thorough Knowledge:** Decision-makers need to be well-versed in safety norms and systems. This includes regular training and updates on safety standards. A study in the International Journal of Environmental Research and Public Health found that effective safety leadership correlates with improved safety performance.

2. Environmental Factors:

- **Unsafe Situational and Climatic Conditions:** These include factors such as excessive noise, high temperatures, and humidity. The World Health Organization (WHO) has identified occupational noise as a significant health risk, leading to hearing loss and increased stress levels.
- **Bad Working Conditions:** Unhealthy environments, like those with poor ventilation or excessive dust and fumes, can lead to chronic respiratory issues. For example, a study by the Indian Journal of Occupational and Environmental Medicine reported high incidences of respiratory problems among workers in poorly ventilated factories.
- **Physical Hazards:** Slippery floors and excessive glare are common causes of workplace accidents. According to the Occupational Safety and Health Administration (OSHA), slips, trips, and falls account for a significant percentage of workplace injuries.
- **Supervisor Behavior:** The behavior of supervisors can also impact safety. Authoritarian and domineering supervisors can create stressful work environments, leading to higher accident rates. The Journal of Occupational Health Psychology found that supportive leadership is linked to better safety outcomes and lower stress levels among employees.

3. Current Examples:

- **COVID-19 Pandemic:** The pandemic has highlighted the need for robust health and safety measures in the workplace. Companies that failed to implement effective safety protocols saw higher transmission rates among employees, leading to severe disruptions and health risks.
- **Heatwaves and Climate Change:** Increasing global temperatures are making it more challenging to maintain safe working conditions, especially in outdoor and non-air-conditioned environments. The International Labour Organization (ILO) estimates that by 2030, over 2% of total working hours worldwide will be lost due to high temperatures, equivalent to 80 million full-time jobs.

In conclusion, a genuine commitment to safety requires action, not just documentation. This includes creating a strong safety culture, ensuring thorough knowledge of safety norms, and addressing environmental factors that contribute to unsafe working conditions.

Industrial accidents have significant consequences that extend beyond immediate physical harm to workers. They also impact industries, workers, consumers, and society at large in various ways. Here's an explanation of each point using current facts:

1. **Loss to Industry:**

- **Medical Treatment Costs:** Industries incur substantial expenses for medical treatment of injured workers. For instance, in the U.S., occupational injuries cost businesses nearly \$60 billion annually in direct costs, which includes medical expenses and lost wages.
- **Wages During Absence:** Companies must continue to pay wages to workers during their recovery period. This is a legal requirement in many countries and adds to the financial burden.
- **Machine and Tool Services:** When an accident occurs, the machinery and tools involved often need inspection and repairs, which adds to operational costs. For example, after a major incident, companies might spend millions on repairing or replacing damaged equipment.
- **Recruitment and Training:** Recruiting and training new workers to replace injured or deceased ones incurs additional costs. The manufacturing sector in particular spends significant amounts on training due to the technical nature of the work.
- **Production Delays and Overtime:** Accidents can halt production, requiring overtime to meet deadlines. This not only increases wages but also impacts productivity and customer satisfaction.
- **Psychological Impact on Workers:** Accidents can cause fear and nervousness among other workers, potentially leading to more accidents. Studies have shown that workplaces with recent accident histories often experience increased stress and anxiety among employees, affecting overall productivity.

2. **Loss to the Workers:**

- **Severe Impact:** Workers face severe consequences, both financially and emotionally. For instance, in countries without strong social safety nets, injured workers may face long-term poverty if they cannot return to work.
- **Family Burden:** The burden on families can be immense, especially if the injured

worker was the primary breadwinner. Families may struggle with medical bills and loss of income, which can lead to long-term financial instability.

- **Permanent Disability:** If a worker is permanently disabled, they might become a burden on their family, requiring long-term care and support. This not only affects the worker but also places emotional and financial strain on the family.

3. Loss to the Consumers:

- **Increased Production Costs:** The costs associated with industrial accidents often get passed down to consumers. This can lead to higher prices for goods and services. For example, following accidents in manufacturing plants, the cost of products like electronics or automobiles might increase.
- **Impact on Living Standards:** Higher prices can reduce consumers' purchasing power, affecting their standard of living. When essential goods become more expensive, it can lead to broader economic challenges, particularly for low-income households.

4. Loss to the Society:

- **Social Burden:** Society often has to step in to support the families of injured or deceased workers. This might include social welfare programs, charitable donations, and other forms of assistance. In many developing countries, the lack of adequate support systems can exacerbate the societal burden.

1. **Economic Impact:** The loss of productive workers can impact the economy. For instance, widespread accidents in critical industries like mining or construction can slow down

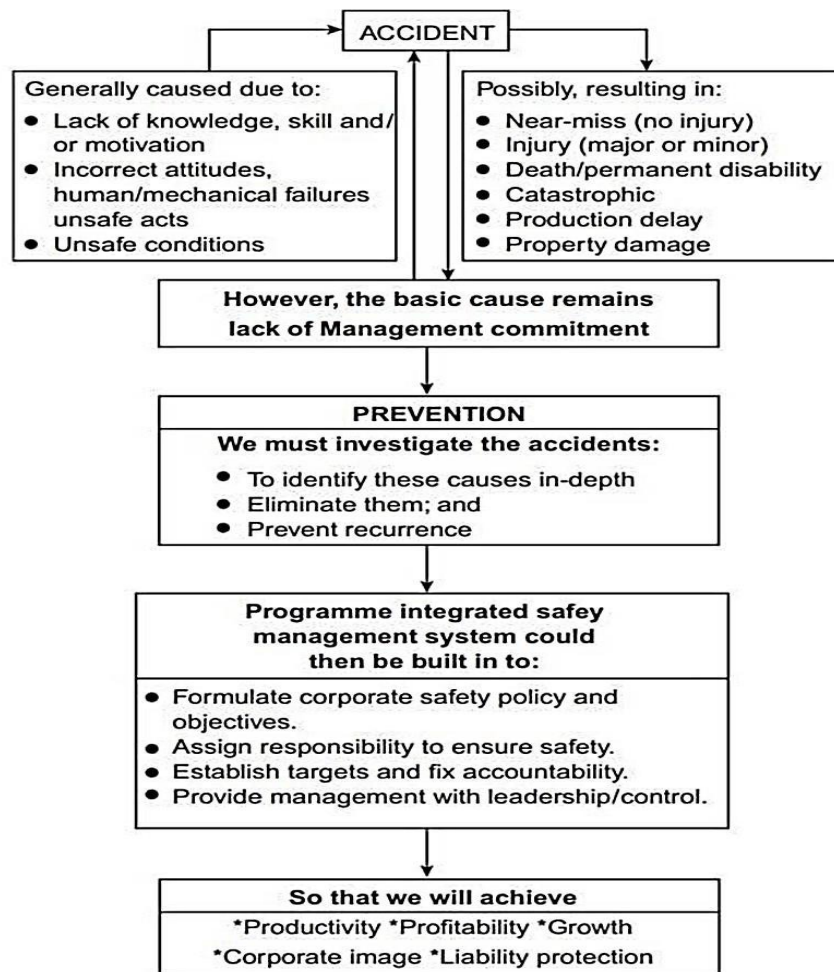
- economic growth, as these sectors are vital to infrastructure and development.

Current Examples:

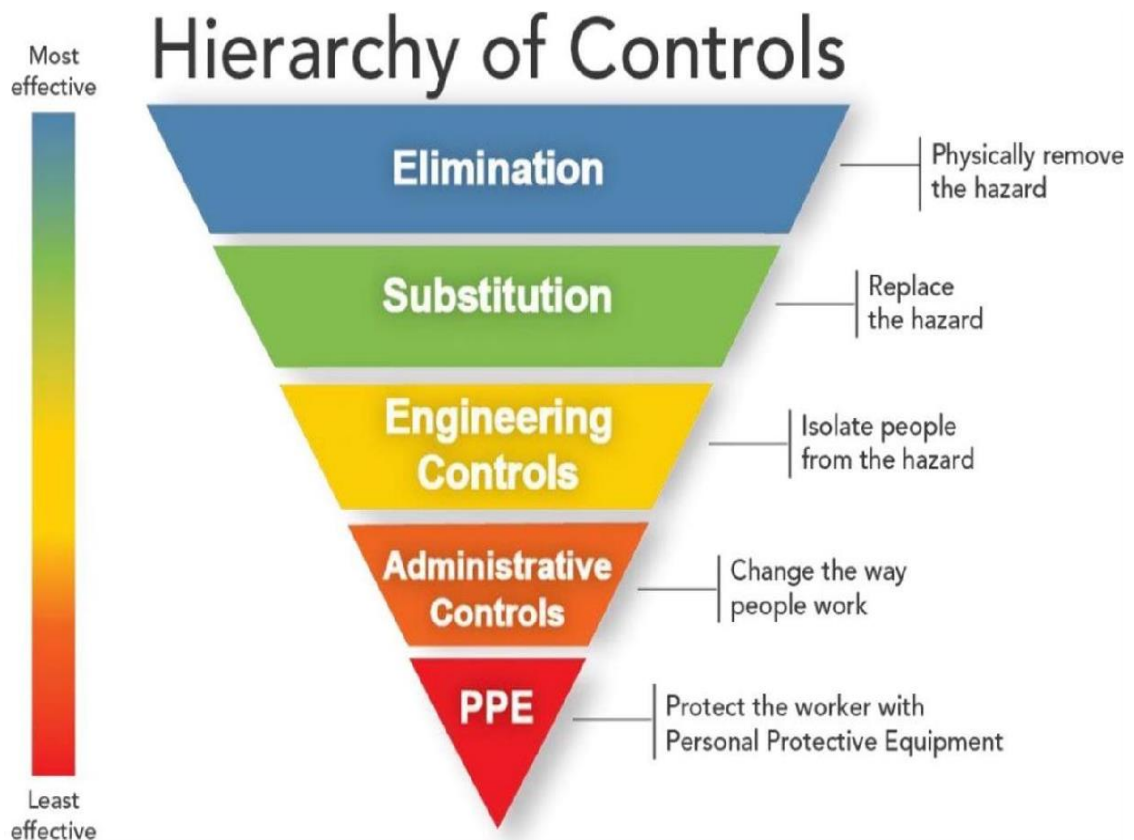
- The 2020 Beirut explosion in Lebanon, caused by improperly stored ammonium nitrate, resulted in massive casualties and economic damage, highlighting the catastrophic impact of industrial accidents on workers, industries, and society.
- In 2021, a chemical plant explosion in Gujarat, India, not only led to immediate deaths and injuries but also had long-term economic impacts on the local community and increased costs for the industry involved.

In summary, industrial accidents have far-reaching consequences that affect

Accident cause and effect SMBO model



Control of Accidents



The hierarchy of controls is a framework used in industrial safety to minimize or eliminate exposure to hazards. It ranks control methods from the most effective to the least effective, guiding safety professionals in implementing the best strategies to protect workers. Here's a breakdown of the hierarchy of controls:

1. Elimination:

- **Description:** This is the most effective control measure. It involves physically removing the hazard from the workplace.
- **Example:** If a particular chemical is hazardous, eliminating its use in the process altogether is the best solution. For instance, replacing a toxic solvent with a non-toxic one.

2. Substitution:

- **Description:** This involves replacing a hazardous substance or process with a less hazardous one.
- **Example:** Using water-based paints instead of solvent-based paints to reduce exposure to harmful fumes.

3. Engineering Controls:

- **Description:** These controls involve isolating people from the hazard. This can be achieved through design changes, equipment modifications, or adding barriers.
- **Example:** Installing ventilation systems to remove hazardous fumes, or machine guards to prevent accidental contact with moving parts.

4. Administrative Controls:

- **Description:** These controls change the way people work. They are less effective than engineering controls because they rely on human behavior and compliance.
- **Example:** Implementing safety training programs, rotating workers to limit exposure time, and establishing standard operating procedures to ensure safe practices.

5. Personal Protective Equipment (PPE):

- **Description:** This is the least effective control measure. It involves providing workers with equipment to protect them from hazards. PPE does not eliminate the hazard but rather minimizes the risk of injury.
- **Example:** Using gloves, safety goggles, helmets, earplugs, and respirators to protect workers from exposure to hazards.

Application of Hierarchy of Controls:

1. Chemical Exposure:

- **Elimination:** Discontinue the use of a hazardous chemical.

- **Substitution:** Replace the hazardous chemical with a safer alternative.
- **Engineering Controls:** Use fume hoods or local exhaust ventilation to capture and remove airborne contaminants.
- **Administrative Controls:** Implement safe handling procedures and provide training on chemical safety.
- **PPE:** Provide gloves, goggles, and respirators to workers handling the chemical.

2. Machine Safety:

- **Elimination:** Design out the need for the hazardous machine.
- **Substitution:** Replace the machine with one that has better safety features.
- **Engineering Controls:** Install guards and safety interlocks.
- **Administrative Controls:** Develop and enforce lockout/tagout procedures, and provide training on safe machine operation.
- **PPE:** Provide workers with safety gloves and protective eyewear.

Mechanical Hazards

Definition:

Mechanical hazards refer to dangers originating from the operation of machinery and mechanical processes in industrial settings. These hazards are typically associated with moving parts, sharp edges, and other physical elements of equipment that can cause harm to workers.

Common Mechanical Injuries:

1. **Cuts and Lacerations:** Caused by sharp edges, cutting tools, or machinery parts.
2. **Crush Injuries:** Result from being caught between moving parts or between a machine and a fixed object.
3. **Amputations:** Severe injury where a limb or extremity is completely severed, often due to machinery without proper guarding.
4. **Fractures:** Bones broken by being hit by or caught in machinery.
5. **Abrasions:** Skin scraped off by rough or moving surfaces.
6. **Punctures:** Deep wounds caused by sharp objects or machinery.
7. **Strains and Sprains:** Injuries to muscles or ligaments due to overexertion or awkward movements while operating machinery.
8. **Burns:** Caused by contact with hot machinery parts or by sparks and friction.

Causes of Mechanical Injuries:

1. **Lack of Guarding:** Missing or inadequate guards around moving parts.
2. **Poor Maintenance:** Faulty or poorly maintained machinery that can malfunction.

3. **Inadequate Training:** Workers not properly trained on safe machine operation and hazard awareness.
4. **Human Error:** Mistakes made by workers due to distraction, fatigue, or lack of experience.
5. **Improper Use:** Using machinery for purposes it wasn't designed for.
6. **Failure to Follow Safety Procedures:** Ignoring lockout/tagout procedures or bypassing safety interlocks.
7. **Loose Clothing and Jewelry:** Can get caught in moving parts.
8. **Unsafe Machine Design:** Machines designed without considering safety aspects.

Prevention of Mechanical Injuries:

1. **Machine Guarding:**
 - Install appropriate guards on all moving parts, cutting tools, and other hazardous machine elements.
 - Ensure guards are always in place and properly maintained.
2. **Regular Maintenance:**
 - Perform routine maintenance to keep machinery in good working order.
 - Replace or repair any faulty parts immediately.
3. **Training and Education:**
 - Provide comprehensive training on machine operation, hazard recognition, and emergency procedures.
 - Conduct regular refresher courses and safety drills.
4. **Safe Work Practices:**
 - Implement and enforce lockout/tagout procedures to ensure machines are de-energized during maintenance.
 - Develop standard operating procedures for safe machine use.
5. **Personal Protective Equipment (PPE):**
 - Provide appropriate PPE such as gloves, safety glasses, face shields, and protective footwear.
 - Ensure workers use PPE correctly and consistently.
6. **Design Improvements:**
 - Opt for machinery with built-in safety features and ergonomic designs.
 - Conduct risk assessments during the purchase and installation of new equipment.
7. **Housekeeping:**
 - Maintain clean and organized work areas to prevent slips, trips, and falls.
 - Keep walkways and machine areas free from obstructions and clutter.
8. **Supervision and Monitoring:**

- Regularly monitor work practices and machine operations to ensure compliance with safety protocols.
- Employ supervisors to oversee safety practices and provide immediate correction when necessary.

9. **Emergency Preparedness:**

- Equip the workplace with first aid kits and ensure quick access to emergency medical services.
- Train workers in first aid and emergency response procedures.

Summary:

Mechanical hazards in industries pose significant risks to workers, leading to various injuries such as cuts, amputations, and fractures. Understanding these hazards and their causes is crucial for implementing effective prevention strategies. These include proper machine guarding, regular maintenance, comprehensive training, and the use of personal protective equipment. By prioritizing safety and adhering to best practices, industries can significantly reduce the occurrence of mechanical injuries and create a safer work environment.

Electrical Hazards in Industries

Definition:

Electrical hazards are dangerous situations where workers are exposed to electrical energy. These hazards can lead to injuries, fatalities, and damage to equipment. They are prevalent in industrial settings due to the extensive use of electrical systems and equipment.

Common Electrical Injuries:

1. **Electrocution:** The most severe injury, leading to death when a person comes into direct contact with a high-voltage source.
2. **Electric Shock:** Occurs when electrical current passes through the body, causing muscle contractions, burns, and potentially fatal injuries.
3. **Burns:** Caused by electrical arcs, hot surfaces, or equipment malfunctioning.
4. **Falls:** Indirectly caused by electrical shocks, leading to loss of balance or coordination.

Causes of Electrical Hazards:

1. **Improper Grounding:**

- Grounding is essential to eliminate unwanted voltage by providing a safe path for

electrical current to the earth. Improper grounding can lead to electrical shocks and equipment damage.

- **Prevention:** Ensure proper grounding of all electrical systems and never remove the ground pin from plugs.

2. **Exposed Electrical Parts:**

- Exposed wires or terminals pose a risk of electric shock or arc flash. Such parts may be found in damaged panels or improperly maintained equipment.
- **Prevention:** Use proper coverings for wires and terminals, ensure all openings in panels are closed, and replace damaged insulation.

3. **Inadequate Wiring:**

- Using undersized or improperly rated wires can cause overheating, leading to fires or equipment failure.
- **Prevention:** Use the correct type and size of wiring for the application, following electrical codes and standards.

4. **Overhead Power Lines:**

- Contact with overhead power lines can result in severe electrical shock or electrocution, especially in construction and maintenance activities.
- **Prevention:** Maintain a safe distance from power lines, use appropriate warning signs, and train workers on the hazards.

5. **Damaged Insulation:**

- Insulation protects conductors and prevents accidental contact. Damaged insulation increases the risk of electrical shock and short circuits.
- **Prevention:** Regularly inspect insulation, replace damaged sections, and avoid using cords with visible wear.

6. **Overloaded Circuits:**

- Overloading occurs when more current flows through a circuit than it is designed to handle, leading to overheating and potential fires.
- **Prevention:** Avoid daisy-chaining multiple devices on a single outlet, use circuit breakers and fuses, and ensure circuits are appropriately rated.

7. **Wet Conditions:**

- Water is a good conductor of electricity, and wet conditions can increase the risk of electric shock.
- **Prevention:** Use Ground Fault Circuit Interrupters (GFCIs), keep electrical equipment dry, and avoid using electrical devices in wet areas.

8. **Damaged Tools and Equipment:**

- Tools and equipment with frayed cords, exposed wires, or damaged parts can cause electric shocks and other injuries.

- **Prevention:** Regularly inspect and maintain tools and equipment, remove damaged items from service, and use tools according to their specifications.

Prevention Strategies:

- **Regular Inspections and Maintenance:** Conduct routine checks of electrical systems, equipment, and tools to identify and fix potential hazards.
- **Training and Awareness:** Provide comprehensive training for employees on electrical safety, hazard recognition, and emergency response procedures.
- **Proper Use of PPE:** Equip workers with appropriate Personal Protective Equipment (PPE) such as insulated gloves, safety goggles, and protective clothing.
- **Compliance with Standards:** Adhere to national and international electrical safety standards and regulations, such as the National Electrical Code (NEC) or OSHA standards.
- **Safety Procedures:** Develop and enforce safety protocols, such as lockout/tagout procedures, to ensure equipment is de-energized before maintenance.

Conclusion:

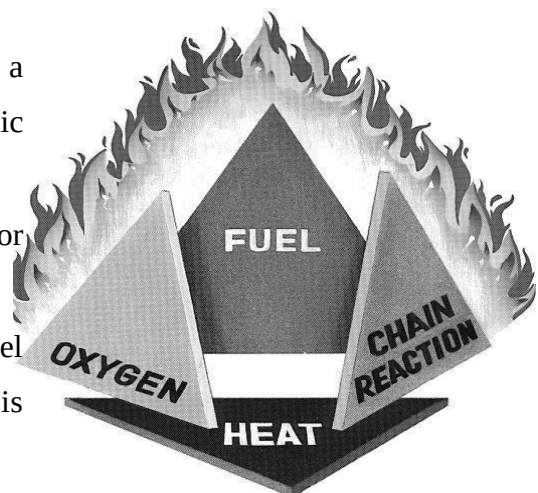
Electrical hazards are a significant concern in industrial environments, posing risks of severe injuries and fatalities. By understanding the common causes and implementing effective prevention strategies, industries can reduce these risks and create a safer work environment for all employees.

Fire Hazards in Industries

Fire:

A state, process, or instance of combustion in which fuel or other material is ignited and combined with oxygen, giving off light, heat, and flame.

- For a fire to occur, fuel, oxygen, heat, and a chemical chain reaction must join in a symbiotic relationship.
- A fire can be classified into two general forms or modes: flame fire and surface fire.
- Flame fires directly burn gaseous or vaporized fuel and include deflagrations. The rate of burning is usually high, and a high temperature is produced.
- Surface fires occur on the surfaces of a solid fuel.



- A surface fire can be represented by a triangle composed of heat, fuel, and air.

**Source: Accident Prevention manual
for Business and Industry by P.E. Hagan ,
J.F. Montgomery and J.T.O'Reilly**

Definition:

Fire hazards in industries refer to situations or conditions that increase the risk of a fire starting, spreading, or causing significant damage. These hazards can arise from various sources, including flammable materials, faulty electrical systems, and inadequate safety protocols. Industrial fires can lead to severe injuries, fatalities, loss of property, and significant economic impact.

Types of Fires and Extinguishers:

Fires are classified into different types based on the fuel source, and each type requires specific types of extinguishers:

1. Class A Fires:

- **Fuel Source:** Ordinary combustibles such as wood, paper, cloth, rubber, and certain plastics.
- **Extinguishers:** Water, foam, or dry chemical extinguishers.

2. Class B Fires:

- **Fuel Source:** Flammable liquids and gases like gasoline, oil, paint, and propane.
- **Extinguishers:** Foam, CO₂, dry chemical extinguishers, or halon.

3. Class C Fires:

- **Fuel Source:** Electrical equipment, including wiring, circuit breakers, machinery, and appliances.
- **Extinguishers:** CO₂, dry chemical, or halon extinguishers. Do not use water or foam extinguishers.

4. Class D Fires:

- **Fuel Source:** Combustible metals such as magnesium, titanium, sodium, and potassium.
- **Extinguishers:** Dry powder extinguishers specifically designed for metal fires.

5. Class K Fires:

- **Fuel Source:** Cooking oils and fats typically found in commercial kitchens.
- **Extinguishers:** Wet chemical extinguishers.

		Ordinary Combustibles	Wood, Paper, Cloth, Etc.
		Flammable Liquids	Grease, Oil, Paint, Solvents
		Live Electrical Equipment	Electrical Panel, Motor, Wiring, Etc.
		Combustible Metal	Magnesium, Aluminum, Etc.
		Commercial Cooking Equipment	Cooking Oils, Animal Fats, Vegetable Oils

TYPE OF FIRE	RECOMMENDED FIRE EXTINGUISHERS				
					
	✓	✗	✓	✓	✓
	α	✓	✓	✗	✗
	✓	✓	✗	✗	✗
	✓	α	α	✗	α
	α	α	α	✓	✗

Source:QRFS(Quick Response Fire Supply)

Source:VirginiaNFPA, USA

Common Fire Injuries:

1. **Burns:** These can range from minor to severe, including first-degree (superficial), second-degree (partial thickness), and third-degree (full thickness) burns.
2. **Smoke Inhalation:** Breathing in smoke can lead to respiratory issues, carbon monoxide poisoning, or asphyxiation.
3. **Injury from Falls or Collisions:** Panic during a fire can cause individuals to trip, fall, or collide with objects, leading to injuries.
4. **Heat Exhaustion:** Prolonged exposure to high temperatures can lead to heat exhaustion or heat stroke.

Causes of Fire Hazards:

1. Flammable and Combustible Materials:

- Improper storage, handling, or disposal of materials like fuels, chemicals, and paper can easily ignite and cause fires.
- **Prevention:** Store flammable materials in designated areas, away from ignition sources, and use appropriate containers.

2. Electrical Hazards:

- Faulty wiring, overloaded circuits, and malfunctioning electrical equipment can lead to electrical fires.
- **Prevention:** Regularly inspect and maintain electrical systems, use equipment according to specifications, and avoid overloading circuits.

3. Heating Equipment:

- Equipment like boilers, heaters, and ovens can be potential sources of fire if not properly maintained.
- **Prevention:** Ensure regular maintenance and inspection of heating equipment, and keep combustible materials away from heat sources.

4. Hot Work Processes:

- Activities such as welding, cutting, and soldering generate sparks and heat that can ignite flammable materials.
- **Prevention:** Implement a hot work permit system, use fire-resistant barriers, and keep fire extinguishers nearby.

5. Human Error:

- Negligence, lack of training, or improper handling of equipment can lead to accidental fires.
- **Prevention:** Provide thorough training on fire safety procedures, ensure proper supervision, and encourage a culture of safety.

6. Chemical Reactions:

- Incompatible chemicals can react violently, producing heat, flames, or toxic gases.
- **Prevention:** Store chemicals according to their compatibility, label containers clearly, and educate workers on chemical hazards.

7. Poor Housekeeping:

- Accumulation of waste, dust, and debris can create fuel for fires.
- **Prevention:** Maintain a clean work environment, regularly dispose of waste, and implement good housekeeping practices.

Prevention Techniques:**1. Fire Risk Assessment:**

- Conduct regular fire risk assessments to identify potential hazards and implement corrective actions.

2. Fire Detection and Alarm Systems:

- Install smoke detectors, heat sensors, and fire alarms to provide early warning of a fire.

3. Fire Suppression Systems:

- Equip buildings with sprinklers, fire extinguishers, and other fire suppression systems.

4. Emergency Evacuation Plans:

- Develop and practice emergency evacuation plans, including designated assembly points and escape routes.

5. Training and Education:

- Provide regular fire safety training to employees, covering topics such as fire prevention, proper use of extinguishers, and emergency response.

6. Proper Storage and Handling of Flammables:

- Store flammable and combustible materials in approved containers and locations, away from ignition sources.

7. Regular Maintenance:

- Perform routine maintenance and inspections on electrical systems, heating equipment, and other potential ignition sources.

8. Housekeeping:

- Maintain a clean and organized workplace, removing waste and debris regularly to reduce fire risks.

9. Hot Work Permits:

- Implement a hot work permit system for tasks involving open flames or sparks, ensuring all safety measures are in place before work begins.

10. Monitoring and Control Systems:

- Use advanced monitoring systems to detect abnormal conditions, such as gas leaks or equipment malfunctions, that could lead to a fire.

Fire Detection:**1. Human Observer**

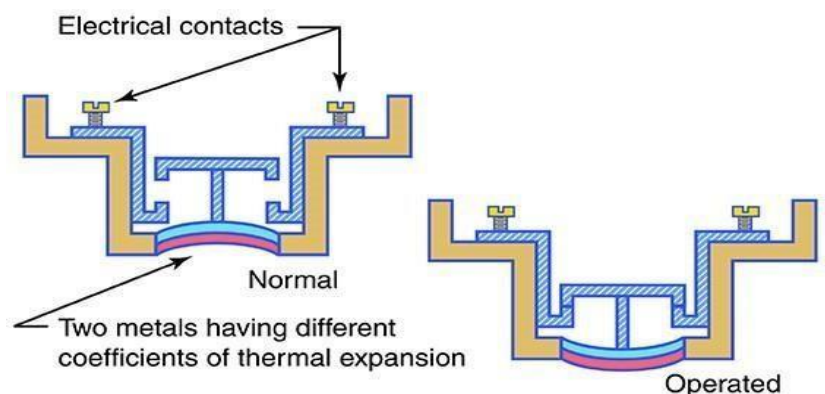
- A human observer is a good fire-detection system for the following reason: He or she can take immediate action in a flexible way, whether calling the fire department or putting out a fire with an extinguisher. Be sure that employees report any fire that they have put out.
- Human observers should also report malfunctioning fire alarm systems. False alarms have a negative effect: if an actual fire occurs, the sounding alarm may be dismissed as just another false alarm.

2. Thermal/Heat Detectors

- Thermal detectors detect the heat from a fire. There are several kinds of thermal detectors, each with a specific use. However, they can only detect the heat of a fire, which usually will not build up to significant levels until the last stage of a fire.
- Many fires start slowly, with little heat generated at the beginning, and will be well under way by the time a thermal detector comes into operation. They are generally used where no life hazard is involved and some loss can be tolerated.

2.1 Fixed-Temperature Detectors

- These thermal detectors are based on a bimetallic element. They are made of two metals that have

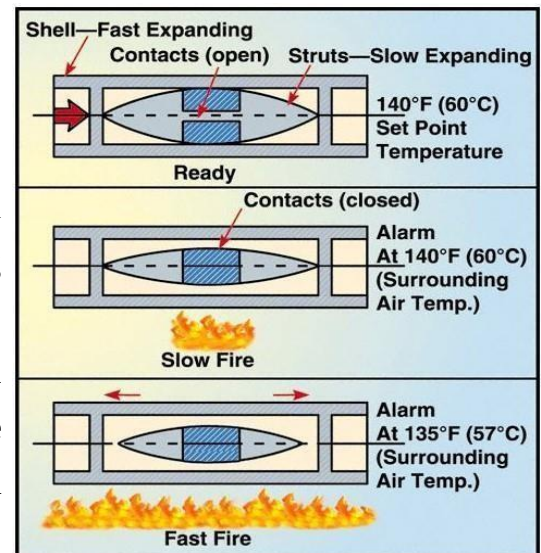


different coefficients of expansion. When heated, the element will bend to close a circuit, initiating the alarm.

- A thermal detector may also use a fusible, spring-loaded element, that melts at a certain temperature, releasing an arm to close a circuit.
- Fixed-temperature detectors are simple, inexpensive, and require a low-voltage.

2.2 Rate-Compensated Thermal Detectors

- These detectors work by the expansion characteristics of a hollow tubular shell containing two curved expansion struts under compression.
- When subjected to a rapid rise of heat, the shell expands and lengthens at a faster rate than the struts, thus permitting the contacts to close. When heated slowly, both the shell and the struts

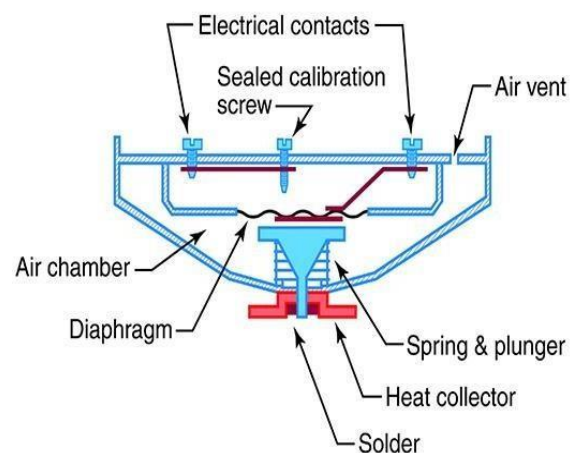


lengthen at about the same rate until the struts are fully extended, thus making contact at a preset point.

2.3 Rate-of-Rise Thermal Detectors

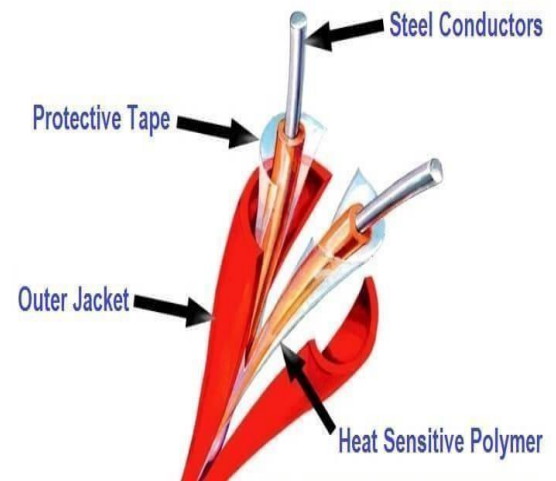
- These detectors use an enclosed, vented hemispherical chamber containing air at atmospheric pressure, with a small pressure-sensitive diaphragm on top. With a normal rise in temperature, the excess pressure is relieved through small vents.
- However, a rapid rise in temperature will deflect the diaphragm faster than the vents can operate, thus triggering an alarm. This unit responds quickly to a rapid rise in temperature.

Rate-of-Rise Heat Detector



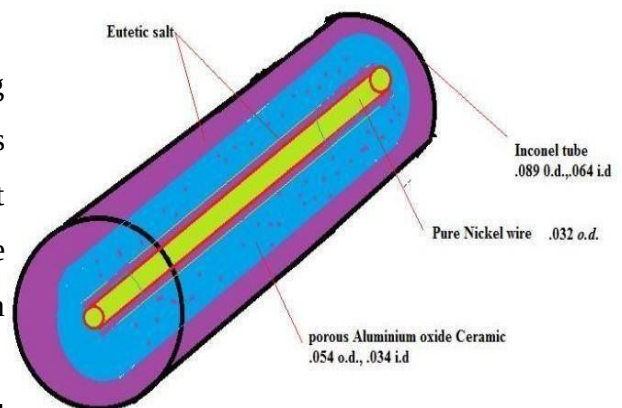
2.4 Line Thermal Detectors

- These detectors use a length of small-diameter tubing that can be as long as 1,000 ft (305 m). When exposed to the heat of a fire, air inside the tube expands, sending a pressure wave to expand a diaphragm at the end, which in turn triggers an alarm.
- This is an unobtrusive, inexpensive detector. For example, it can be run along the ceiling molding, where it is nearly invisible. No maintenance is needed. It can be painted over, and it will even work with the tubing broken, if the temperature rises fast enough.



2.5 Eutectic-Salt-Line Thermal Detectors

- These detectors consist of pliant metal tubing containing a eutectic salt in which a wire is embedded. At a preselected temperature, the salt creates a short-circuit between the internal wire and the outside tubing, thereby triggering an alarm.
- The pliant tubing can be wound around and shaped to the various components of the engines to signal any increase in temperature that might be the result of a fire caused by leaking oil, hydraulic fluid, etc.
- These line detectors are also used in conveyor-belt systems where the bearings supporting a rubberized belt may ignite because of friction and lack of lubrication.

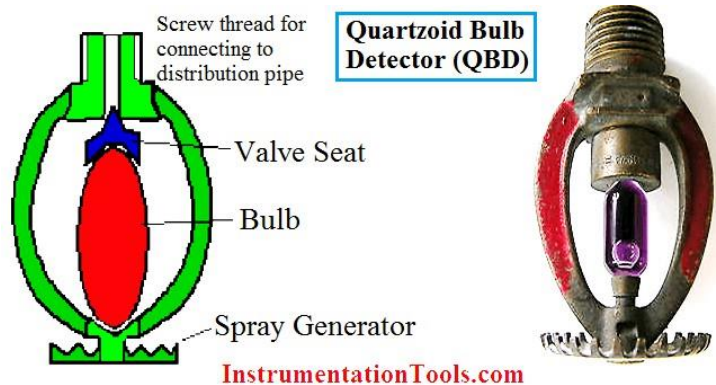


2.6 Bulb Detection Systems

Bulb detection systems are mechanical fire detection devices designed to operate without the need for electrical power, making them suitable for environments where electrical sparking could pose an additional fire risk. These systems are ideal for locations with explosive fire hazards where avoiding the use of electricity is critical.

Key Points:**Mechanical Nature**

- **No Electricity Required:** Bulb detection systems are entirely mechanical, which eliminates the risk of electrical sparks that could ignite flammable materials. This feature makes them particularly valuable in explosive or highly flammable environments.
- **Reliability:** The mechanical design ensures that the system remains operational even in conditions where electrical systems might fail, such as during power outages or in highly combustible areas.

**System Design**

- **Bulbs with Atmospheric Pressure Air:** The system includes a series of bulbs filled with air at atmospheric pressure. These bulbs are strategically placed along the ceiling of the area that needs protection.
- **Diaphragm Connection:** All bulbs are connected via tubing to a diaphragm located at a central control center. The diaphragm acts as the detection mechanism for the system.
- **Heat Detection:** When a fire causes the ambient temperature to rise, the air inside the bulbs expands. This expansion increases the pressure within the bulb and connected tubing.
- **Diaphragm Deflection:** The increased pressure deflects the diaphragm at the control center. This mechanical movement triggers the fire suppression system.

Activation Mechanism

- **Temperature Rise Response:** The system responds to a rise in temperature that affects one or more of the bulbs, causing the diaphragm to deflect.
- **Mechanical Extinguishing System:** Upon diaphragm deflection, a mechanical fire extinguishing system is activated. This system can release fire suppression agents such as CO₂.

- **Applications:** Bulb detection systems are commonly used as a release mechanism for CO₂ fire extinguishers, especially in marine and industrial applications where electrical systems might pose additional risks.

Applications

- **Marine Environments:** In ships and marine environments, where fire hazards are significant and electrical systems might be compromised by water or humidity, bulb detection systems offer a reliable fire detection and suppression solution.
- **Industrial Settings:** In industrial settings with high fire risk, such as chemical plants, oil refineries, and explosive manufacturing facilities, bulb detection systems provide a safe and effective means of detecting and responding to fires without the need for electrical power.

Advantages

- **Safety in Hazardous Areas:** By eliminating the use of electricity, these systems reduce the risk of igniting volatile substances.
- **Reliability and Simplicity:** The mechanical nature of the system ensures it remains functional even in harsh conditions where electronic systems might fail.
- **Cost-Effective:** These systems are generally cost-effective to install and maintain compared to complex electronic fire detection systems.

Smoke Detectors: Detailed Explanation

Smoke detectors are crucial for early fire detection. Each type of smoke detector employs different technology and principles to identify the presence of smoke, offering unique advantages and applications.

Beam Photoelectric Detectors

- **Principle:** Utilize a light beam to detect smoke.
- **Components:** Consist of a light source (transmitter) and a light sensor (receiver) installed opposite each other.

How They Work:

- **Normal Operation:** In the absence of smoke, the light beam travels unobstructed from the transmitter to the receiver.
- **Smoke Detection:** When smoke interrupts the beam, the light scatters, reducing the intensity that reaches the receiver. This change in light intensity activates the alarm.

Advantages:

- **Coverage Area:** Capable of monitoring large areas.
- **Sensitivity:** Highly sensitive to smoke, providing early detection.

Applications:

- **Large Open Spaces:** Suitable for warehouses, atriums, and gymnasiums where installing multiple point detectors would be impractical.
- Reflected Beam Photoelectric Detectors
- **Principle:** Similar to beam photoelectric detectors but use a single unit with a reflector.

How They Work:

- **Normal Operation:** The combined transmitter and receiver unit emits a light beam towards the reflector, which bounces the beam back to the receiver.
- **Smoke Detection:** Smoke in the beam path reduces the reflected light intensity, triggering the alarm.

Advantages:

- **Simplified Installation:** Only one device needs to be mounted and wired.
- **Flexibility:** Easier to install in areas with challenging layouts.

Applications:

- **Complex Layouts:** Ideal for spaces where installing separate transmitter and receiver units is difficult.

Products of Combustion (Ionization) Detectors

- **Principle:** Use a small amount of radioactive material to ionize the air in a sensing chamber.

How They Work:

- **Normal Operation:** Ionized air allows a steady electrical current to flow between two electrodes.
- **Smoke Detection:** Smoke particles disrupt the ionization process, reducing the current flow and triggering the alarm.

Advantages:

- **Rapid Response:** Quick to detect fast-flaming fires with small smoke particles.
- **Cost-Effective:** Generally less expensive than photoelectric detectors.

Applications:

- **Residential and Commercial Buildings:** Effective for environments prone to fast-flaming fires.

Single Chamber Ionization Detectors

- **Design:** Feature a single ionization chamber.

How They Work:

- **Normal Operation:** Ionized air in the chamber allows a current to flow.
- **Smoke Detection:** Smoke particles entering the chamber disrupt the ionization process, reducing the current and activating the alarm.

Advantages:

- **Simple Design:** Easy to install and maintain.
- **Affordable:** Cost-effective option for basic smoke detection needs.

Applications:

- **Homes and Small Offices:** Commonly used in residential settings and smaller commercial spaces.

Dual Chamber Ionization Detectors

- **Design:** Have two ionization chambers – one open to the air and the other sealed.

How They Work:

- **Normal Operation:** The chambers provide a baseline comparison of ionized air.
- **Smoke Detection:** Smoke entering the open chamber disrupts the ionization, creating a difference between the chambers that triggers the alarm.

Advantages:

- **Enhanced Sensitivity:** More accurate detection compared to single chamber detectors.
- **False Alarm Reduction:** Better at differentiating between actual fires and non-threatening disturbances.

Applications:

- **Mixed Environments:** Suitable for areas where both fast-flaming and smoldering fires might occur.

Low Voltage Ionization Detectors

- **Design:** Operate at lower voltage levels compared to standard ionization detectors.

How They Work:

- **Normal Operation:** Utilize lower voltage to ionize the air in the chamber.
- **Smoke Detection:** Smoke particles disrupt the ionization process at low voltage, triggering the alarm.

Advantages:

- **Energy Efficiency:** Lower power consumption makes them ideal for battery-operated systems.
- **Compact Size:** Can be smaller in size due to lower power requirements.

Applications:

- **Battery-Powered Systems:** Ideal for use in portable and standalone smoke detection systems.

Conclusion

Choosing the right smoke detector depends on various factors, including the environment, type of potential fire hazards, and installation challenges. Here's a summary:

- **Beam Photoelectric Detectors:** Best for large open areas.
- **Reflected Beam Photoelectric Detectors:** Suitable for complex layouts and easier installation.
- **Products of Combustion (Ionization) Detectors:** Effective for fast-flaming fires.
- **Single Chamber Ionization Detectors:** Cost-effective for basic needs.
- **Dual Chamber Ionization Detectors:** Enhanced sensitivity and reduced false alarms.
- **Low Voltage Ionization Detectors:** Energy-efficient and compact.

Flame Detectors:

Flame detectors are essential in environments where the risk of fire is high. They detect the presence of flame using various technologies and are commonly used in industrial settings, refineries, and other high-risk areas. Here's a detailed look at different types of flame detectors:

Infrared (IR) Flame Detectors

Overview:

- **Principle:** Detect infrared radiation emitted by flames.
- **Components:** Consist of sensors that detect IR radiation in specific wavelengths.

How They Work:

- **Normal Operation:** The detector monitors IR radiation in the environment.
- **Flame Detection:** Flames emit IR radiation at characteristic wavelengths. When the detector senses this radiation, it triggers an alarm.

Advantages:

- **Sensitivity:** Can detect flames through smoke, dust, and steam.
- **Speed:** Provides quick detection of flames.

Applications:

- **Industrial Settings:** Used in refineries, chemical plants, and storage facilities.
- **Outdoor Environments:** Suitable for open areas where other detection methods might fail.

Limitations:

- **False Alarms:** Can be triggered by other IR sources, like hot machinery.

Ultraviolet (UV) Flame Detectors

Overview:

- **Principle:** Detect ultraviolet radiation emitted by flames.
- **Components:** Use UV sensors sensitive to wavelengths emitted by flames.

How They Work:

- **Normal Operation:** The detector monitors UV radiation in the environment.
- **Flame Detection:** Flames emit UV radiation at specific wavelengths. When the detector senses this radiation, it triggers an alarm.

Advantages:

- **Speed:** Extremely fast detection of flames.
- **Selective Detection:** Less likely to be triggered by sunlight or other IR sources.

Applications:

- **Indoor Settings:** Effective in enclosed spaces such as turbine enclosures and generator rooms.
- **Laboratories:** Used in environments where quick detection is critical.

Limitations:

- **Limited Range:** UV radiation is easily absorbed by smoke and other particulates.

Infrared/Ultraviolet (IR/UV) Flame Detectors Overview:

- **Principle:** Combine both IR and UV detection technologies.
- **Components:** Use both IR and UV sensors to detect flames.

How They Work:

- **Normal Operation:** The detector monitors both IR and UV radiation.
- **Flame Detection:** Detects the simultaneous presence of IR and UV radiation, which is characteristic of flames, to trigger an alarm.

Advantages:

- **Reduced False Alarms:** The combination of technologies reduces the likelihood of false alarms.
- **Improved Reliability:** Provides a higher level of detection accuracy.

Applications:

- **High-Risk Environments:** Ideal for oil and gas industries, petrochemical plants, and hazardous material storage.

Limitations:

- **Cost:** More expensive than single-technology detectors.

Multi-Spectrum Infrared (MSIR) Flame Detectors**Overview:**

- **Principle:** Detects flames using multiple IR wavelengths.
- **Components:** Use multiple IR sensors tuned to different wavelengths.

How They Work:

- **Normal Operation:** The detector monitors multiple IR wavelengths.
- **Flame Detection:** Detects characteristic IR signatures of flames across several wavelengths, increasing accuracy and reducing false alarms.

Advantages:

- **High Accuracy:** Enhanced discrimination between flames and other IR sources.
- **Robust Performance:** Effective in challenging environments with smoke, dust, and steam.

Applications:

- **Complex Environments:** Used in industrial plants, power generation facilities, and aircraft hangars.

Limitations:

- **Complexity:** More complex and expensive to install and maintain.

Visible Flame Detectors Overview:

- **Principle:** Detect visible light emitted by flames.
- **Components:** Use sensors sensitive to visible light.

How They Work:

- **Normal Operation:** The detector monitors visible light in the environment.
- **Flame Detection:** Detects the presence of visible light flickering characteristic of flames to trigger an alarm.

Advantages:

- **Simplicity:** Straightforward technology, easy to understand and deploy.
- **Cost:** Generally less expensive than advanced IR or UV detectors.

Applications:

- **Controlled Environments:** Suitable for areas where visible flames are expected and other sources of light are controlled.

Limitations:

- **False Alarms:** Can be triggered by other sources of visible light.

Comparison of Flame Detectors

Type of Detector	Detection Method	Speed	Accuracy	False Alarms	Applications
Infrared (IR)	IR Radiation	Fast	Moderate	Prone to IR sources	Refineries, chemical plants, outdoor areas
Ultraviolet (UV)	UV Radiation	Very Fast	High	Less prone to sunlight	Indoor settings, laboratories
Infrared/Ultraviolet (IR/UV)	Combined IR and UV	Fast	High	Reduced	Oil and gas, hazardous material storage
Multi-Spectrum Infrared (MSIR)	Multiple IR Wavelengths	Fast	Very High	Reduced	Industrial plants, aircraft hangars
Visible Flame	Visible Light	Moderate	Moderate	Prone to visible light	Controlled environments

Fire Alarm Systems

Fire alarm systems are critical for detecting fires early and alerting occupants to evacuate. They can also notify emergency services, helping to mitigate damage and ensure safety. Here are the main types of fire alarm systems:

1. Local Alarm Systems

Overview:

- Local alarm systems are designed to alert the occupants of a building to the presence of a fire.

Components:

- Detectors:** Smoke, heat, or flame detectors that sense fire.
- Alarms:** Bells, sirens, or flashing lights that notify occupants.
- Control Panel:** Manages the signals from detectors and triggers alarms.

Operation:

- Detection:** When a detector senses a fire, it sends a signal to the control panel.
- Alarm Activation:** The control panel activates the alarms, alerting occupants.
- No External Notification:** These systems do not automatically notify emergency services.

Advantages:

- **Simplicity:** Easy to install and maintain.
- **Cost-Effective:** Generally less expensive than more complex systems.
- **Immediate Notification:** Quickly alerts building occupants.

Limitations:

- **No External Alert:** Requires manual notification of emergency services.
- **Limited to Building:** Only alerts those within the building.

Applications:

- **Small Buildings:** Suitable for small offices, retail spaces, and residential buildings.

2. Auxiliary Alarm Systems

Overview:

- Auxiliary alarm systems are designed to provide local alerts and also notify emergency services through a connection to a municipal fire alarm system.

Components:

- **Detectors:** Smoke, heat, or flame detectors.
- **Alarms:** Bells, sirens, or flashing lights.
- **Control Panel:** Manages signals and alarm activation.
- **Municipal Connection:** A direct line to a municipal fire alarm system or a fire department.

Operation:

- **Detection:** A detector senses a fire and sends a signal to the control panel.
- **Alarm Activation:** The control panel activates local alarms.
- **External Notification:** The system sends a signal to the municipal fire alarm system, notifying emergency services.

Advantages:

- **Dual Notification:** Alerts building occupants and emergency services simultaneously.
- **Improved Response:** Faster emergency response times due to automatic notification.

Limitations:

- **Installation Complexity:** Requires a connection to a municipal system.
- **Higher Cost:** More expensive than local alarm systems.

Applications:

- **Medium to Large Buildings:** Suitable for schools, commercial buildings, and industrial facilities.

3. Central Station Systems

Overview:

- Central station systems are monitored by an external central monitoring station, providing continuous oversight and immediate emergency notification.

Components:

- **Detectors:** Smoke, heat, or flame detectors.
- **Alarms:** Bells, sirens, or flashing lights.
- **Control Panel:** Manages signals and alarm activation.
- **Communication Link:** A connection to a central monitoring station, often via phone lines or the internet.

Operation:

- **Detection:** A detector senses a fire and sends a signal to the control panel.
- **Alarm Activation:** The control panel activates local alarms.
- **External Notification:** The control panel sends a signal to the central monitoring station.
- **Monitoring:** The monitoring station assesses the alarm and contacts emergency services if necessary.

Advantages:

- **Continuous Monitoring:** 24/7 oversight ensures quick detection and response.
- **Professional Management:** Central station staff handle emergency notifications and coordination.
- **Enhanced Security:** Often includes monitoring for other security breaches.

Limitations:

- **Cost:** Higher installation and ongoing monitoring fees.
- **Dependence on Communication:** Requires reliable communication links.

Applications:

- **Large and High-Risk Buildings:** Suitable for hospitals, large commercial buildings, and high-rise buildings.

Comparison of Fire Alarm Systems

System Type	Notification Method	External Notification	Monitoring	Cost	Applications
Local Alarm Systems	Alarms alert building occupants	No	No	Low	Small buildings, offices, homes
Auxiliary Alarm Systems	Local alarms and municipal connection	Yes	No	Moderate	Medium to large buildings, schools

Central Station Systems	Local alarms and central monitoring	Yes	Yes (24/7)	High	Large buildings, hospitals, high-rises
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Fire Extinguishing

1. Portable Fire Extinguishers

- The term portable is applied to manual equipment used on small, beginning fires or used between the discovery of a fire and the functioning of automatic equipment or the arrival of professional fire fighters.
- Portable extinguishers are classified to indicate their ability to handle specific classes and sizes of fires.
- **Use Class A extinguishers** for ordinary combustibles, such as wood, paper, some plastics, and textiles, where a quenching-cooling effect is required.
- **Use Class B extinguishers** for flammable liquid and gas fires, such as oil, gasoline, paint, and grease fires, where oxygen exclusion or a flame interrupting effect is essential.
- **Use Class C extinguishers** for fires involving electrical wiring and equipment where the dielectric non conductivity of the extinguishing agent is of first importance. These units are not classified by a numeral, because Class C fires are essentially either Class A or Class B but also involve energized electrical wiring and equipment. Therefore, choose the coverage of the extinguisher for the burning fuel.
- **Use Class D extinguishers** for fires in combustible metals, such as magnesium, potassium, powdered aluminum, zinc, sodium, titanium, zirconium, and lithium. Persons working in areas where Class D fire hazards exist must be aware of the dangers in using Class A, B, or C extinguishers on a Class D fire. Of course, they should also know the correct way to extinguish Class D fires. These units are not classified by a numerical system and are intended for special hazard protection only.

Types of Fire Extinguishers

1. **Water (APW) Extinguishers**
2. **Foam Extinguishers**
3. **Carbon Dioxide (CO2) Extinguishers**
4. **Dry Chemical Extinguishers**
5. **Wet Chemical Extinguishers**
6. **Class D Extinguishers**
7. **Class K Extinguishers**

1. Water (APW) Extinguishers

Class of Fire:

- **Class A:** Ordinary combustibles (wood, paper, cloth, trash, plastics)

Mechanism:

- APW stands for "Air-Pressurized Water."
- Uses water to cool down the burning material, effectively reducing the heat of the flames.

Advantages:

- Simple and easy to use.
- Effective for common combustibles.
- Non-toxic and inexpensive.

Disadvantages:

- **Not suitable for:** Class B (flammable liquids) and Class C (electrical) fires, as water can spread flammable liquids and conduct electricity.
-

2. Foam Extinguishers

Class of Fire:

- **Class A:** Ordinary combustibles
- **Class B:** Flammable liquids (oil, gasoline, paints, solvents)

Mechanism:

- Forms a blanket over the fire, smothering it and preventing re-ignition.
- The foam cools the fire and seals the flammable vapors.

Advantages:

- Effective on both Class A and B fires.
- Prevents re-ignition by forming a barrier between the fuel and the oxygen.

Disadvantages:

- **Not suitable for:** Class C (electrical) fires, as foam can conduct electricity.
-

3. Carbon Dioxide (CO2) Extinguishers

Class of Fire:

- **Class B:** Flammable liquids
- **Class C:** Electrical fires

Mechanism:

- Displaces oxygen, smothering the fire.
- CO2 is cold as it exits the extinguisher, which helps to cool the fire.

Advantages:

- Leaves no residue.

- Effective on electrical fires without causing damage to equipment.

Disadvantages:

- **Not suitable for:** Class A fires, as it may not displace enough oxygen to effectively cool down the fire.
-

4. Dry Chemical Extinguishers

Class of Fire:

- **Class A, B, and C** (multipurpose dry chemical)
- **Class B and C** (ordinary dry chemical)

Mechanism:

- Uses a chemical powder to interrupt the chemical reaction of the fire.
- Multipurpose dry chemical extinguishers often use monoammonium phosphate.

Advantages:

- Versatile, can be used on multiple types of fires.
- Provides rapid extinguishing.

Disadvantages:

- Leaves a residue that can be harmful to sensitive equipment.
 - May obscure vision temporarily during discharge.
-

5. Wet Chemical Extinguishers

Class of Fire:

- **Class K:** Cooking oils and fats

Mechanism:

- Forms a soapy foam on the surface of the burning oil or fat, preventing re-ignition.
- Contains a special potassium-based solution that reacts with the burning oils/fats to create a layer of foam.

Advantages:

- Highly effective on deep fat fryers and other cooking appliances.
- Cools and smothers the fire simultaneously.

Disadvantages:

- **Not suitable for:** Class A, B, C, or D fires.
-

6. Class D Extinguishers

Class of Fire:

- **Class D:** Flammable metals (magnesium, titanium, sodium, etc.)

Mechanism:

- Uses a dry powder that absorbs heat and forms a crust to smother the fire.

Advantages:

- Specifically designed for metal fires.
- Prevents the metal from reacting with air or water.

Disadvantages:

- **Not suitable for:** Class A, B, C, or K fires.
-

7. Class K Extinguishers

Class of Fire:

- **Class K:** Cooking oils and fats

Mechanism:

- Uses a wet chemical agent that reacts with the cooking fat or oil to form a foam layer.
- This process, called saponification, cools the fire and prevents re-ignition.

Advantages:

- Specifically designed for high-temperature cooking oil fires.
- Effective in commercial kitchens.

Disadvantages:

- **Not suitable for:** Class A, B, C, or D fires.
-

Fire Extinguisher Classes and Symbols

- **Class A:** Green triangle with "A" inside - Ordinary combustibles.
- **Class B:** Red square with "B" inside - Flammable liquids.
- **Class C:** Blue circle with "C" inside - Electrical equipment.
- **Class D:** Yellow star with "D" inside - Flammable metals.
- **Class K:** Black hexagon with "K" inside - Cooking oils and fats.

Sprinkler and Water-Spray Systems

- There are many types of sprinklers and water-spray systems for extinguishing fires. The type of building, operations performed in it, and materials used will help determine the type of sprinkler or water-spray system used. Automatic sprinklers are the most extensively used and most effective installations of fixed fire-extinguishing systems.
- There are six basic types of automatic sprinkler systems: wet-pipe, dry-pipe, preaction, deluge, combined dry-pipe and preaction, and limited water-supply systems.
- In the **wet-pipe system**, which represents the greatest percentage of sprinkler installations, all parts of the system's piping are filled up to the sprinkler heads with water under pressure. Then, when heat actuates the sprinkler, water is immediately sprayed over the area below.

- The **dry-pipe system** generally substitutes for a wet-pipe system in areas where piping is exposed to freezing temperatures. A good rule of thumb is to use a dry-pipe system when more than 20 sprinklers are involved.
- **Preaction systems** are similar to dry-pipe systems. However, they react faster and hence minimize water damage in case of fire or mechanical damage to sprinklers or piping.
- The **deluge system** wets down an entire area by admitting water to sprinklers that are open at all times. Deluge valves that control the water supply to the system are actuated by a fire-detection system located in the same area as the sprinklers. This type of system is primarily designed for extra hazard buildings where great quantities of water may have to be applied immediately over large areas.
- The combination dry-pipe and preaction systems are used on installations that are larger than can be accommodated by one dry-pipe valve.
- The limited water-supply system is used for installations that do not have access to a continual or large supply of water.

Water-Spray Systems

- Water spray is effective on all types of fires where there is no hazardous chemical reaction between the water and the materials that are burning.
- Water-spray systems can be designed effectively for any one, or any combination, of the following purposes:
 - extinguishing fire
 - controlling fire where extinguishment is not desirable, such as gas leaks
 - exposure protection; that is, absorbing heat transferred from equipment by the spray
 - preventing fire by having water spray dissolve, dilute, disperse, or cool flammable materials.

Dry-chemical piped systems

- have been developed for situations where quick extinguishment is needed, either in a confined area or for localized application, and where reignition is unlikely. These systems are adaptable to flammable liquid and electrical hazards. They can be operated manually or automatically, or be activated at the system or by remote control.
- Automatic or manually controlled steam jet systems can be used to smother fires in closed containers or in small rooms, such as heaters, drying kilns, smoke ovens, asphalt-mixing tanks, and dry-cleaning tumbler dryers. However, such systems are practical only where a large supply of steam is continuously available.
- However, steam has not been found effective on deep-seated fires that may form glowing embers, or in enclosures where the normal operating temperature is not considerably higher than air temperatures.

Inert gas systems

- It can prevent fires and explosions by replacing the oxygen in the air with an inert gas, such as CO₂, nitrogen, flue gas, or other non-combustible gases, until it reaches a level (or percentage) where combustion will not take place.
- Preventing the development of explosive mixtures is the best defense against explosions. Equipment for handling and storing flammable gases should be designed, constructed, inspected, and maintained so that the danger of leakage and of explosive mixture formation is reduced to a minimum.

Explosion suppression system

- It can be used to reduce the destructive pressure of an explosion. These systems are designed to detect an explosion as it is starting and to actuate devices that suppress, vent, or take other action to prevent the full explosive force.

Safety Colour Codes

Safety colour codes play a crucial role in fire safety, providing visual cues to quickly identify hazards, emergency equipment, and safe practices. These colour codes are standardized to ensure consistency and immediate recognition. Here's an explanation of the primary safety color codes used in fire safety:

1. **Red**
2. **Yellow**
3. **Green**
4. **Blue**
5. **Orange**
6. **Black and White**

1. Red

Use:

- **Fire Protection Equipment and Apparatus**

Examples:

- Fire extinguishers
- Fire alarm boxes
- Fire hoses and cabinets
- Emergency stop buttons

Meaning:

- Indicates the presence of fire-related hazards or equipment.
- Immediate action may be required in the event of a fire.

Key Points:

- Red is universally recognized as the color for danger and emergency situations.
 - Ensures quick identification of fire fighting tools and emergency systems.
-

2. Yellow**Use:**

- **Caution**

Examples:

- Hazardous areas
- Warning signs
- Caution tape
- Slip, trip, and fall hazards

Meaning:

- Indicates general warning and the need for caution.
- Alerts individuals to potential hazards that could cause injury.

Key Points:

- Yellow is used to draw attention to risks that may not be immediately obvious.
 - Often used in conjunction with black stripes for added visibility.
-

3. Green**Use:**

- **Safety Equipment and Information**

Examples:

- Emergency exit signs
- First aid stations
- Safety equipment locations
- Safe routes and egress pathways

Meaning:

- Indicates safe conditions, locations of safety equipment, and emergency exits.
- Signals areas where safety measures and equipment are available.

Key Points:

- Green is associated with safety and the presence of life-saving equipment.
 - Helps in identifying safety procedures and locations during an emergency.
-

4. Blue

Use:

- **Mandatory Actions**

Examples:

- Signs indicating specific actions that must be followed (e.g., "Wear Personal Protective Equipment")
- Information signs
- Safety instructions

Meaning:

- Indicates that a specific behavior or action is required.
- Provides safety instructions and information that must be adhered to.

Key Points:

- Blue signals the need for compliance with safety procedures and guidelines.
 - Ensures that necessary actions are taken to maintain safety.
-

5. Orange

Use:

- **Warning of Dangerous Parts of Machinery and Equipment**

Examples:

- Guards and moving parts of machinery
- High-voltage areas
- Hot surfaces

Meaning:

- Warns of hazardous parts of machinery that could cause injury.
- Indicates areas where there is a potential for burns, cuts, or electrical hazards.

Key Points:

- Orange highlights the presence of potentially dangerous equipment or components.
 - Prompts awareness and caution when working near these areas.
-

6. Black and White

Use:

- **Housekeeping and Traffic Areas**

Examples:

- Directional signs
- Housekeeping markings
- Areas that require routine maintenance

- Boundaries of walkways and traffic lanes

Meaning:

- Used to indicate housekeeping and organizational details.
- Guides the flow of traffic and demarcates different areas within a facility.

Key Points:

- Black and white provide clear, contrasting visuals for non-hazardous areas.
- Helps maintain order and organization in the workplace.

Trucks, rail cards and other containers must have placards attached indicating the hazard level of the container's contents.



Salient Points of the India Factories Act, 1948 for Health and Safety

The Factories Act, 1948, was established to regulate labor in factories and ensure the safety, health, and welfare of workers. The formulation of this act was driven by historical developments, social changes, and the need for industrial regulation in India. Here's a detailed explanation of how the Factories Act, 1948, was made:

Historical Context

1. Early Industrialization:

- During the 19th century, India saw the advent of industrialization, particularly with the establishment of textile mills and other manufacturing industries.
- This period was marked by poor working conditions, long hours, and the exploitation of labor, including women and children.

2. Initial Regulations:

- The first attempt to regulate factory conditions in India was the Factories Act of 1881, which focused primarily on child labor and basic safety measures.
- Subsequent amendments were made in 1891, 1911, 1922, and 1934, each broadening the scope of regulations to include more comprehensive safety, health, and welfare provisions.

Need for a Comprehensive Law

1. Post-World War II Industrial Growth:

- After World War II, there was significant industrial growth in India, necessitating more robust laws to address the complex and varied challenges of modern industrial establishments.

2. Social and Political Changes:

- The nationalist movement and the push for social reforms highlighted the need for better labor laws to protect workers' rights and ensure their welfare.
- There was growing awareness and pressure from labor unions and social reformers to enact comprehensive legislation that would provide adequate safeguards for workers.

Formulation Process

1. Government Committees and Reports:

- The Government of India set up various committees and commissions to study labor conditions and recommend necessary legislation.
- One significant report was the Royal Commission on Labour in India (1931), which provided detailed insights and recommendations on labor conditions in Indian factories.

2. Drafting the Legislation:

- Based on these recommendations and subsequent studies, a draft bill was prepared by the government.
- The draft bill went through several stages of scrutiny, including reviews by labor experts, industrialists, and representatives from labor unions.

3. Consultations and Debates:

- Extensive consultations and debates were held in the Central Legislative Assembly (the legislative body at the time) to refine the provisions of the bill.
- Input was sought from various stakeholders, including factory owners, labor representatives, and social reformers, to ensure a balanced and effective law.

Enactment

1. Introduction and Passing:

- The Factories Bill was introduced in the Central Legislative Assembly in 1947.
- After thorough discussions, debates, and revisions, the bill was passed by the Assembly.

2. Assent and Implementation:

- The Factories Act, 1948, received the assent of the Governor-General of India on September 23, 1948.
- The Act came into force on April 1, 1949, marking a significant milestone in labor legislation in India.

Key Provisions

The Factories Act, 1948, covered various aspects, including:

- **Health:** Cleanliness, disposal of wastes, ventilation, and temperature control.
- **Safety:** Fencing of machinery, work on or near machinery in motion, and employment of young persons on dangerous machines.
- **Welfare:** Washing facilities, facilities for storing and drying clothing, sitting arrangements,

first-aid appliances, and canteens.

- **Working Hours:** Regulation of working hours for adults, employment of young persons, and provisions for overtime.
- **Annual Leave:** Provisions for annual leave with wages.
- **Special Provisions:** Measures for hazardous processes and factories employing women and children.

Amendments

- The Act has undergone several amendments to address emerging industrial challenges and to ensure alignment with international labor standards. These amendments have further strengthened the provisions related to health, safety, and welfare of workers.

The Factories Act, 1948 is a comprehensive legislation aimed at ensuring the health, safety, and welfare of workers in factories. Here are the salient points concerning health and safety, wash rooms, drinking water, lighting, cleanliness, fire safety, guarding, pressure vessels, and safety color codes:

Health and Safety

1. **General Duties of Occupier and Manufacturer (Section 7A, 7B):**
 - Ensure the health, safety, and welfare of all workers.
 - Provide information, instruction, training, and supervision.
 - Ensure safe use, handling, storage, and transport of articles and substances.
 - Maintain all places of work in a condition that is safe and without risks to health.
2. **Fencing of Machinery (Section 21):**
 - Every moving part of a prime mover and every flywheel connected to a prime mover shall be securely fenced.
3. **Work on or Near Machinery in Motion (Section 22):**
 - Only trained adult male workers wearing tight-fitting clothing are allowed to work on or near machinery in motion.
4. **Employment of Young Persons on Dangerous Machines (Section 23):**
 - No young person shall work at any machine that is dangerous unless they have been fully instructed and supervised.
5. **Prohibition of Employment of Women and Children Near Cotton Openers (Section 27):**
 - No woman or child shall be employed in any part of a factory where a cotton opener is at work.

Wash Rooms**1. Provision of Wash Rooms (Section 42):**

- Separate and adequately screened washing facilities for male and female workers.
- Facilities should be conveniently accessible and maintained in a clean and orderly condition.

Drinking Water**1. Drinking Water (Section 18):**

- Sufficient supply of wholesome drinking water shall be provided and maintained at suitable points conveniently accessible to all workers.
- Such points should be marked with "Drinking Water" in a language understood by the majority of workers.

Lighting**1. Lighting (Section 17):**

- Sufficient and suitable lighting, natural or artificial, shall be provided in every part of the factory where workers are working or passing.

Cleanliness**1. Cleanliness (Section 11):**

- Factory premises to be kept clean and free from dirt and refuse.
- Effective arrangements for the disposal of waste and effluents.

Fire Safety**1. Precautions in Case of Fire (Section 38):**

- All practicable measures to be taken to prevent the outbreak of fire and its spread, both internally and externally.
- Safe means of escape for all workers in case of fire.
- Necessary equipment for extinguishing fire to be provided and maintained.

Guarding**1. Safety Guarding of Machinery (Sections 24-28):**

- Guarding of all dangerous parts of machinery.
- Prohibition of cleaning, lubricating, or adjusting machinery while in motion.

Pressure Vessels**1. Pressure Plant (Section 31):**

- Pressure plants shall be examined and tested by a competent person periodically.
- Report of such examination and test to be kept available for inspection.

Safety Color Codes

1. Safety Color Codes (Schedule III):

- Color codes are used to denote different types of hazards and safety instructions:
 - **Red:** Fire protection equipment and danger.
 - **Yellow:** Caution and physical hazards.
 - **Green:** Safety and first-aid equipment.
 - **Blue:** Information and caution (mainly for non-immediate hazards).

Chapter 3 of the Factories Act, 1948

Chapter 3 of the Factories Act, 1948, is dedicated to the health of workers in factories. This chapter lays down detailed provisions to ensure the maintenance of a healthy working environment. Here is a detailed explanation of the sections related to health under this chapter:

Section 11: Cleanliness

- Factories must be kept clean and free from effluvia arising from any drain, privy, or other nuisance.
- Accumulations of dirt and refuse must be removed daily by sweeping or by any other effective method.
- Effective measures must be taken to ensure that the floors of every workroom are cleaned at least once every week by washing, using disinfectant, or by some other effective method.

Section 12: Disposal of Wastes and Effluents

- Effective arrangements must be made in every factory for the treatment of wastes and effluents due to the manufacturing process carried out in the factory.
- These arrangements must conform to the rules made by the State Government in this regard.

Section 13: Ventilation and Temperature

- Adequate ventilation by the circulation of fresh air and suitable temperature must be maintained in each workroom to ensure that the workers do not suffer from discomfort or injury to health.
- This must be done in such a manner that it prevents the formation of high humidity in the workrooms.

Section 14: Dust and Fume

- Effective measures must be taken to prevent the inhalation and accumulation of dust and fumes in workrooms.
- Exhaust appliances must be provided to remove dust, fumes, or other impurities from the workplace.

Section 15: Artificial Humidification

- If artificial humidification is employed, the State Government may make rules to ensure proper conditions.
- These rules may include standards of purity of water used for humidification, measures for controlling humidity, and maintenance of records.

Section 16: Overcrowding

- No room in any factory shall be overcrowded to an extent injurious to the health of the workers employed therein.
- The floor space for every worker should be at least 14.2 cubic meters in case of factories in existence before the commencement of this Act and 9.9 cubic meters for factories established after its commencement.

Section 17: Lighting

- Sufficient and suitable lighting, natural or artificial, or both, must be provided and maintained in every part of a factory where workers are working or passing.
- All glazed windows and skylights used for the lighting of the workroom must be kept clean on both the inner and outer surfaces and free from obstructions.

Section 18: Drinking Water

- A sufficient supply of wholesome drinking water must be provided and maintained at suitable points conveniently situated for all workers.
- The State Government may make rules for the examination of the quality of the water and the method of supplying it.

Section 19: Latrines and Urinals

- Sufficient latrine and urinal accommodation must be provided, maintained, and kept clean in

every factory.

- These facilities must be conveniently situated and accessible to workers at all times while they are in the factory.
- Separate and adequately screened accommodation must be provided for male and female workers.

Section 20: Spittoons

- In every factory, there must be provided a sufficient number of spittoons in convenient places, and they must be maintained in a clean and hygienic condition.
- No worker shall spit within the premises of the factory except in the spittoons provided for the purpose.
- A notice containing this provision and the penalty for its infringement must be prominently displayed at suitable places in the premises.

These provisions under Chapter 3 of the Factories Act, 1948, aim to create a safe and healthy working environment for factory workers, ensuring their well-being and productivity.

Chapter IV: Safety - The Factories Act, 1948

Chapter IV of the Factories Act, 1948 focuses on the safety of workers in the factory environment. Below are the explanations of the sections related to the pressure plant and precautions in case of fire.

Section 31: Pressure Plant

- **Definition and Scope:** This section applies to plants operated at pressures above atmospheric pressure. This includes steam boilers, pressure vessels, and other equipment.
- **Safety Measures:**
 - Pressure plants must be maintained in such a manner as to ensure their safe working.
 - Regular inspections and maintenance must be conducted to prevent accidents due to pressure.
 - Safety valves and pressure gauges must be installed and maintained in working order.
- **Records and Reports:**
 - Detailed records of inspections and maintenance should be kept.
 - Any defects found must be reported immediately and remedial actions must be taken promptly.

- **Training and Supervision:**

- Workers operating pressure plants must be adequately trained.
- Proper supervision must be ensured to prevent any mishandling.

Section 38: Precautions in Case of Fire

- **Fire Fighting Equipment:**

- Factories must be equipped with suitable fire-fighting equipment, such as fire extinguishers, hydrants, and fire hoses.
- Fire-fighting equipment must be regularly inspected and maintained to ensure they are in working order.

- **Emergency Exits:**

- Factories must have clearly marked emergency exits that are easily accessible and free from obstructions at all times.
- Workers must be informed about the location and use of emergency exits.

- **Fire Drills and Training:**

- Regular fire drills must be conducted to ensure that workers are aware of the fire safety procedures.
- Workers must be trained in using fire-fighting equipment and in evacuation procedures.

- **Fire Alarms:**

- Factories must be equipped with a reliable fire alarm system to alert workers in case of a fire.
- The alarm system must be regularly tested to ensure its effectiveness.

- **Storage of Flammable Materials:**

- Flammable materials must be stored in a safe manner, away from sources of ignition.
- Proper storage facilities must be provided to minimize the risk of fire.

- **Preventive Measures:**

- Smoking should be prohibited in areas where flammable materials are stored or used.
- Electrical installations should be regularly inspected to prevent short circuits or sparks.

By adhering to these safety measures, factories can significantly reduce the risk of accidents and ensure a safer working environment for their employees.

Unit: II: MAINTENANCE ENGINEERING

Maintenance Engineering

1. Definition and Aim of Maintenance Engineering

Definition: Maintenance Engineering involves the application of engineering principles and practices to the optimal operation, upkeep, and repair of machinery and equipment. The field ensures that all physical assets in an industrial setup, such as machines, tools, and infrastructure, are maintained in a condition where they function effectively and efficiently throughout their service life.

Aim:

- **Maximize Equipment Availability:** Ensuring that machinery and equipment are always operational when needed.
- **Extend Equipment Life:** Delaying the degradation of equipment by preventing failures and optimizing performance.
- **Enhance Safety and Compliance:** Ensuring that all equipment complies with safety regulations to protect workers and the environment.
- **Reduce Operating Costs:** Minimizing the cost of maintenance by preventing unexpected breakdowns and optimizing the maintenance schedule.
- **Optimize Performance:** Keeping equipment running at peak performance levels to maintain product quality and production efficiency.

2. Primary and Secondary Functions and Responsibilities of the Maintenance

Department Primary Functions:

1. **Preventive Maintenance:** Regular, planned maintenance activities to prevent unexpected breakdowns and extend the lifespan of equipment.
 - Scheduled inspections
 - Lubrication
 - Parts replacement before failure
2. **Corrective Maintenance:** Reactive maintenance performed after a fault is detected, to restore the equipment to operational condition.
 - Troubleshooting
 - Repair work

- Emergency response to breakdowns
- 3. **Predictive Maintenance:** The use of data and analytics to predict equipment failures before they occur, enabling maintenance to be performed just in time.
 - Vibration analysis
 - Thermography
 - Oil analysis

Secondary Functions:

1. **Inventory Management:** Ensuring that all necessary spare parts and tools are available when needed, without overstocking or causing excessive costs.
 2. **Documentation:** Keeping accurate records of all maintenance activities, equipment histories, and work orders for audit and future planning.
 3. **Training and Development:** Providing ongoing training to maintenance personnel to ensure they are skilled in the latest maintenance techniques and technologies.
 4. **Safety Management:** Implementing safety protocols and ensuring compliance with safety standards during all maintenance activities.
-

3. Types of Maintenance

1. Preventive Maintenance (PM):

- **Definition:** Scheduled maintenance activities carried out to prevent equipment failures.
- **Examples:** Regular oil changes, filter replacements, and scheduled inspections.
- **Advantages:** Reduces the likelihood of unexpected failures, extends equipment life.
- **Disadvantages:** Can result in unnecessary maintenance if not properly scheduled.

2. Corrective Maintenance (CM):

- **Definition:** Maintenance performed after a fault or failure has occurred.
- **Examples:** Repairing a broken part, fixing leaks, or replacing worn-out components.
- **Advantages:** Focuses on actual issues that need repair.
- **Disadvantages:** Can lead to unexpected downtime and higher repair costs.

3. Predictive Maintenance (PdM):

- **Definition:** Maintenance based on the condition of equipment as monitored by sensors and data analytics.
- **Examples:** Replacing bearings when vibration analysis indicates excessive wear.
- **Advantages:** Minimizes downtime and maintenance costs by performing

maintenance only when necessary.

- **Disadvantages:** Requires investment in monitoring technology and expertise in data analysis.

4. Reliability-Centered Maintenance (RCM):

- **Definition:** A systematic approach that ensures maintenance is done to preserve system function, balancing PM, CM, and PdM.
- **Examples:** A mix of strategies based on the criticality of the equipment.
- **Advantages:** Optimizes maintenance resources and ensures high reliability of critical systems.
- **Disadvantages:** Complex to implement and manage.

5. Condition-Based Maintenance (CBM):

- **Definition:** Maintenance is performed when certain indicators show signs of decreasing performance or impending failure.
 - **Examples:** Maintenance triggered by sensor data indicating higher-than-normal operating temperatures.
 - **Advantages:** Reduces unnecessary maintenance and increases equipment life.
 - **Disadvantages:** Requires accurate and reliable monitoring systems.
-

4. Types and Applications of Tools Used for Maintenance Hand

Tools:

- **Wrenches:** Used for tightening or loosening bolts and nuts.
- **Screwdrivers:** For turning screws to fasten or remove components.
- **Pliers:** Used for gripping, bending, or cutting wires and small objects.
- **Hammers:** For driving nails or adjusting the position of parts.

Power Tools:

- **Drills:** Used for making holes in materials such as wood, metal, or plastic.
- **Grinders:** For cutting, grinding, or polishing surfaces.
- **Impact Wrenches:** Deliver high torque output to loosen or tighten bolts.
- **Saws:** Used to cut materials to size, including metal, wood, and plastic.

Measuring Tools:

- **Calipers:** Precision measuring tools for checking the dimensions of parts.

- **Multimeters:** For checking electrical parameters like voltage, current, and resistance.
- **Thermometers:** To measure the temperature of equipment components.
- **Pressure Gauges:** For checking the pressure within systems, such as hydraulics or pneumatics.

Diagnostic Tools:

- **Vibration Analyzers:** Used to detect imbalances or misalignments in rotating machinery.
 - **Thermographic Cameras:** For detecting hot spots that indicate electrical or mechanical problems.
 - **Ultrasonic Detectors:** To detect leaks or mechanical wear by picking up high-frequency sounds.
 - **Oil Analysis Kits:** To check the condition of lubricants and detect wear particles.
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5. Maintenance Cost and Its Relation with Replacement Economy

Maintenance Cost:

- **Direct Costs:** Labor, spare parts, and materials required for maintenance activities.
- **Indirect Costs:** Downtime, reduced production capacity, and energy inefficiencies caused by equipment not being in optimal condition.
- **Overhead Costs:** Administrative and logistical expenses related to maintenance operations.

Relation with Replacement Economy:

- **Economic Life of Equipment:** The point at which the total cost of maintaining the equipment exceeds the cost of replacement.
 - **Cost-Benefit Analysis:** Regular analysis should be conducted to determine if continued maintenance is more cost-effective than replacing equipment.
 - **Depreciation:** The loss in value of equipment over time, influencing the decision to maintain or replace.
 - **Optimal Replacement Time:** Determined by comparing maintenance costs with the costs of new equipment, factoring in potential productivity gains from newer technology.
-

6. Service Life of Equipment

Definition:

- **Service Life:** The period during which equipment remains functional and meets its

performance standards before it becomes uneconomical to repair.

Factors Affecting Service Life:

- **Usage:** How often and under what conditions the equipment is used (e.g., continuous operation, harsh environments).
- **Maintenance Quality:** Regular and effective maintenance can extend the service life of equipment.
- **Technology Advancements:** Newer technologies may render older equipment obsolete, reducing its effective service life.
- **Environmental Conditions:** Exposure to extreme temperatures, moisture, chemicals, and other environmental factors can shorten equipment life.

Determining Service Life:

- **Manufacturer's Guidelines:** Recommendations provided by equipment manufacturers.
- **Historical Data:** Past performance and failure rates of similar equipment.
- **Condition Monitoring:** Using predictive maintenance tools to monitor the equipment's condition and estimate its remaining service life.

Unit: III: WEAR AND CORROSION AND THEIR PREVENTION

1. Wear

1.1 Definition of Wear

- **Wear** refers to the gradual removal of material from solid surfaces due to mechanical action such as friction, abrasion, or erosion. It is a common problem in machinery and components, leading to the deterioration of performance and eventual failure.
-

1.2 Types of Wear

1. Adhesive Wear:

- **Cause:** Occurs when two solid surfaces slide over each other, causing material transfer from one surface to another.
- **Effect:** Can lead to increased friction, heat, and material loss.
- **Example:** Scuffing in gears.

2. Abrasive Wear:

- **Cause:** Caused by hard particles or hard protuberances forced against and moving along a solid surface.
- **Effect:** Scratching and gouging of the surface, leading to material removal.
- **Example:** Sand particles causing wear in mining equipment.

3. Erosive Wear:

- **Cause:** Occurs when a fluid or gas containing solid particles strikes a surface at high velocity.
- **Effect:** Gradual removal of material from the surface.
- **Example:** Wear on turbine blades due to high-speed water or air particles.

4. Fatigue Wear:

- **Cause:** Repeated cyclic loading leading to material failure after many cycles.
- **Effect:** Surface cracking and material spalling.
- **Example:** Pitting in bearings and gear teeth.

5. Corrosive Wear:

- **Cause:** Occurs when chemical reactions cause the material to degrade, which is then

removed by mechanical action.

- **Effect:** Accelerated wear due to the combined effects of corrosion and mechanical forces.
 - **Example:** Rusting of steel components in marine environments.
-

1.3 Causes of Wear

- **Mechanical Contact:** Sliding, rolling, or impact forces causing material to be worn away.
 - **Surface Roughness:** Irregularities on the surface increase friction and promote wear.
 - **Contaminants:** Particles, dust, or dirt can act as abrasives, causing surface damage.
 - **Improper Lubrication:** Inadequate lubrication increases friction, leading to higher wear rates.
 - **Environmental Factors:** Exposure to harsh environments (e.g., moisture, chemicals) can accelerate wear.
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1.4 Effects of Wear

- **Loss of Material:** Leads to dimensional changes, affecting the fit and function of parts.
 - **Increased Friction:** Causes additional energy consumption and heat generation.
 - **Noise and Vibration:** As surfaces degrade, they can produce unwanted noise and vibration.
 - **Decreased Efficiency:** Worn parts may not perform as intended, leading to reduced operational efficiency.
 - **Potential Failure:** If not addressed, wear can lead to catastrophic failure of components.
-

1.5 Wear Reduction Methods

1. Surface Hardening:

- **Process:** Techniques like carburizing, nitriding, or induction hardening increase surface hardness, making it more resistant to wear.
- **Application:** Gears, shafts, and cutting tools.

2. Use of Wear-Resistant Materials:

- **Materials:** Tungsten carbide, ceramics, and hardened steel are commonly used to resist wear.

- **Application:** Cutting tools, mining equipment, and abrasives.
 - 3. **Improved Surface Finish:**
 - **Process:** Polishing or grinding to reduce surface roughness, decreasing friction and wear.
 - **Application:** Bearings, seals, and sliding components.
 - 4. **Lubrication:**
 - **Role:** Reduces friction and wear by providing a protective film between contact surfaces.
 - **Application:** Bearings, gears, and sliding mechanisms.
 - 5. **Coatings:**
 - **Types:** Hard coatings like chrome, nickel, or PVD (Physical Vapor Deposition) can protect against wear.
 - **Application:** Automotive parts, cutting tools, and aerospace components.
-

1.6 LUBRICANTS - TYPES AND APPLICATIONS

1. **Oil:**
 - **Properties:** Provides a thin film to separate surfaces, reducing friction and wear.
 - **Application:** Engines, hydraulic systems, and gearboxes.
 2. **Grease:**
 - **Properties:** Thicker than oil, provides longer-lasting lubrication, especially in low-speed, high-load applications.
 - **Application:** Bearings, chassis, and machinery joints.
 3. **Solid Lubricants:**
 - **Examples:** Graphite, molybdenum disulfide.
 - **Application:** High-temperature environments, dry conditions, and vacuum applications.
 4. **Synthetic Lubricants:**
 - **Properties:** Designed to provide superior performance in extreme conditions.
 - **Application:** High-performance engines, aerospace applications, and refrigeration systems.
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1.7 LUBRICATION METHODS

1. Screw Down Grease Cup:

- **Working:** A grease cup with a screw-down plunger forces grease into the lubrication point.
- **Application:** Used in older machinery and equipment requiring manual lubrication.

2. Pressure Grease Gun:

- **Working:** Uses a hand pump or compressed air to force grease into lubrication points.
- **Application:** Bearings, fittings, and joints.

3. Splash Lubrication:

- **Working:** The moving parts splash oil around the interior of a casing, lubricating all parts.
- **Application:** Small engines, gearboxes.

4. Gravity Lubrication:

- **Working:** Oil is stored in a reservoir and fed by gravity through a drip feed to the lubrication point.
- **Application:** Steam engines, stationary engines.

5. Wick Feed Lubrication:

- **Working:** A wick draws oil from a reservoir to the lubrication point through capillary action.
- **Application:** Clocks, small machines, and instruments.

6. Side Feed Lubrication:

- **Working:** Oil is fed into the lubrication point from the side, often using a small reservoir or oil cup.
- **Application:** Light machinery, machine tools.

7. Ring Lubrication:

- **Working:** A ring rides on a rotating shaft and carries oil from a reservoir to the lubrication point.
 - **Application:** Bearings, electric motors.
-

2. Corrosion

2.1 Definition and Principle of Corrosion

Definition:

- **Corrosion** is the deterioration of a material, usually a metal, due to a chemical reaction with its environment. The most common form of corrosion is the oxidation of metals, such as the rusting of iron.

Principle:

- Corrosion typically occurs through an electrochemical process, where an anodic reaction (metal loss) and a cathodic reaction (usually oxygen reduction) take place. The presence of an electrolyte, such as water, facilitates this process.
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2.2 Factors Affecting Corrosion

1. **Material Composition:**

- Different metals and alloys have varying degrees of susceptibility to corrosion.

2. **Environmental Conditions:**

- Humidity, temperature, pH, and the presence of salts or chemicals can accelerate corrosion.

3. **Electrochemical Potential:**

- The difference in electrochemical potential between different areas on a metal surface can lead to galvanic corrosion.

4. **Surface Finish:**

- Rough or scratched surfaces are more prone to corrosion than smooth, polished surfaces.

5. **Presence of Oxygen:**

- Oxygen is a key component in many corrosion processes, such as rusting.
-

2.3 Types of Corrosion

1. Uniform Corrosion:

- **Description:** Occurs evenly across the surface of the metal.
- **Example:** General rusting of steel structures.

2. Pitting Corrosion:

- **Description:** Localized corrosion leading to the formation of small pits or holes.
- **Example:** Stainless steel in chloride environments.

3. Galvanic Corrosion:

- **Description:** Occurs when two different metals are in electrical contact in the presence of an electrolyte.
- **Example:** Corrosion of aluminum in contact with copper.

4. Crevice Corrosion:

- **Description:** Localized corrosion in confined spaces where access to the working fluid is restricted.
- **Example:** Corrosion in gasket joints or under washers.

5. Stress Corrosion Cracking (SCC):

- **Description:** The combined effect of tensile stress and a corrosive environment leading to cracking.
- **Example:** Cracking of stainless steel in chloride environments.

6. Intergranular Corrosion:

- **Description:** Corrosion along the grain boundaries of a metal.
- **Example:** Sensitization of stainless steel in the heat-affected zone of welds.

7. Erosion-Corrosion:

- **Description:** Accelerated corrosion due to the relative movement between a corrosive fluid and the metal surface.
 - **Example:** Corrosion of pipes in high-velocity water flow.
-

2.4 Corrosion Prevention Methods

1. Material Selection:

- **Use of Corrosion-Resistant Alloys:** Stainless steel, aluminum, and titanium resist corrosion better than other metals.
- **Coatings:** Applying protective coatings like paint, galvanizing, or anodizing to protect the metal surface from the environment.

2. Cathodic Protection:

- **Sacrificial Anode:** A more reactive metal (like zinc) is used to protect the main metal by corroding preferentially.
- **Impressed Current:** An external current is applied to make the metal act as a cathode, reducing its tendency to corrode.

3. Environmental Control:

- **Reducing Exposure:** Limiting the metal's exposure to corrosive environments, such as by using dehumidifiers or inhibitors.
- **pH Control:** Maintaining the pH at levels that minimize corrosion.

4. Design Improvements:

- **Avoiding Crevices:** Designing components to avoid crevices where corrosive agents can accumulate.
- **Proper Drainage:** Ensuring that water does not accumulate on surfaces.
- **Surface Treatments:** Polishing surfaces to reduce corrosion initiation sites.

5. Corrosion Inhibitors:

- **Chemical Additives:** Adding chemicals to the environment that slow down the corrosion process.
- **Application:** Used in cooling systems, pipelines, and boilers.

6. Regular Maintenance:

- **Inspection and Cleaning:** Regularly inspecting and cleaning components to remove corrosive agents.
- **Reapplying Coatings:** Reapplying protective coatings as needed to maintain protection.

Unit: IV: FAULT TRACING

1. Introduction to Fault Tracing

Definition:

Fault tracing is the systematic approach to diagnosing and identifying the root cause of faults or malfunctions in machinery and equipment. It involves analyzing symptoms, identifying probable causes, and testing hypotheses until the problem is accurately pinpointed.

Importance of Fault Tracing:

- **Reduces Downtime:** Quick identification of issues minimizes downtime and improves operational efficiency.
 - **Enhances Safety:** Proper fault tracing ensures that hidden issues are resolved, maintaining a safe environment for operators.
 - **Cost-Effective:** Prevents damage to equipment by catching faults early, which reduces repair costs and extends equipment life.
 - **Increases Reliability:** Helps ensure that machinery and equipment operate reliably and within performance standards.
-

2. Decision Tree Concept

Definition:

A decision tree is a logical diagram used to systematically track the potential causes of faults in equipment. It starts from the problem at hand, branching out into a series of questions or tests that guide you through a sequence to diagnose the issue.

Importance in Fault Tracing:

- **Structured Approach:** Provides a clear, logical path for diagnosing problems.
- **Ease of Use:** Helps operators follow a standardized approach to troubleshoot faults.
- **Enhanced Accuracy:** Reduces the likelihood of missing critical diagnostic steps.

Applications of Decision Trees in Fault Tracing:

- **Machine Tools:** Used to diagnose mechanical issues like spindle misalignment or feed mechanism faults.
 - **Hydraulic Systems:** Helps in identifying issues such as pump malfunctions or fluid contamination.
 - **Pneumatic Systems:** Useful for diagnosing issues with air leaks, valve malfunctions, or pressure loss.
 - **Automotive Systems:** Aids in tracing faults in fuel, ignition, or exhaust systems.
 - **Thermal Systems:** Used for boilers, identifying issues such as temperature imbalances or blockages.
 - **Electrical Equipment:** Assists in diagnosing motor malfunctions, power supply issues, or wiring faults.
-

3. Sequence of Fault-Finding Activities

The process of fault finding follows a structured sequence:

1. **Problem Identification:**
 - Clearly identify and document the symptoms of the fault.
 - Gather information from operators, maintenance logs, and past service records.
2. **Initial Inspection:**
 - Conduct a visual inspection of the equipment for any obvious signs of malfunction (e.g., leaks, broken components).
 - Check for any unusual sounds, vibrations, or overheating.
3. **Formulating Hypotheses:**
 - Based on the symptoms, list possible causes of the fault.
 - Use a decision tree to organize and test hypotheses systematically.
4. **Testing Hypotheses:**
 - Perform tests or checks to validate or rule out each hypothesis.
 - Use diagnostic tools such as multimeters, pressure gauges, and thermal imaging if needed.
5. **Root Cause Analysis:**
 - Once the fault is identified, determine the root cause (e.g., lack of lubrication, worn components).
 - Check if other components may have been affected.
6. **Documentation and Reporting:**

- Document the findings, the root cause, and the corrective action taken.
 - Update maintenance records to aid in future fault tracing.
-

4. Decision Trees for Various Equipment Example: Decision Tree for Faults in Machine

Tools Problem: Machine tool is not operating correctly.

- **Step 1:** Check power supply.
 - *Yes:* Move to next step.
 - *No:* Restore power supply.
- **Step 2:** Check for loose or misaligned components.
 - *Yes:* Realign or tighten components.
 - *No:* Move to next step.
- **Step 3:** Check lubrication levels.
 - *Yes:* Move to next step.
 - *No:* Add lubrication.
- **Step 4:** Check for wear on moving parts.
 - *Yes:* Replace or repair worn parts.
 - *No:* Continue inspection based on symptoms.

Example: Decision Tree for Hydraulic Systems

Problem: Hydraulic system is not providing adequate pressure.

- **Step 1:** Check hydraulic fluid levels.
 - *Yes:* Move to next step.
 - *No:* Refill hydraulic fluid.
- **Step 2:** Inspect for leaks in the hydraulic line.
 - *Yes:* Repair or replace the leaking section.
 - *No:* Move to next step.
- **Step 3:** Check the pump for proper operation.
 - *Yes:* Move to next step.
 - *No:* Repair or replace the pump.
- **Step 4:** Check for clogs in filters or valves.
 - *Yes:* Clean or replace clogged components.
 - *No:* Continue diagnosis based on pressure readings.

Example: Decision Tree for Air Compressors

Problem: Air compressor is not reaching desired pressure.

- **Step 1:** Check air intake filters.
 - Yes: Clean or replace filters.
 - No: Move to next step.
- **Step 2:** Inspect for leaks in the compressed air line.
 - Yes: Repair or replace leaking sections.
 - No: Move to next step.
- **Step 3:** Check pressure relief valve settings.
 - Yes: Adjust valve settings.
 - No: Move to next step.
- **Step 4:** Inspect the motor or drive belt.
 - Yes: Repair or replace worn components.
 - No: Continue diagnosis based on compressor performance.

Example: Decision Tree for Internal Combustion Engines Problem:

Engine fails to start.

- **Step 1:** Check fuel supply.
 - Yes: Move to next step.
 - No: Add fuel.
- **Step 2:** Check the battery and electrical connections.
 - Yes: Move to next step.
 - No: Charge or replace the battery.
- **Step 3:** Check spark plugs or ignition system.
 - Yes: Replace or clean spark plugs.
 - No: Move to next step.
- **Step 4:** Inspect air intake for blockages.
 - Yes: Clean the air intake.
 - No: Continue diagnosis based on engine performance.

Example: Decision Tree for Boilers Problem:

Boiler pressure is not maintained.

- **Step 1:** Check water level in the boiler.
 - Yes: Move to next step.
 - No: Fill the boiler with water.
- **Step 2:** Inspect for leaks in the boiler and piping.
 - Yes: Repair leaks.
 - No: Move to next step.
- **Step 3:** Check fuel supply and burner operation.
 - Yes: Adjust burner settings or fuel flow.
 - No: Move to next step.
- **Step 4:** Inspect safety and pressure relief valves.
 - Yes: Replace faulty valves.
 - No: Continue diagnosis based on pressure levels.

Example: Decision Tree for Electrical Motors Problem:

Motor fails to start.

- **Step 1:** Check power supply and connections.
 - Yes: Move to next step.
 - No: Restore power supply.
- **Step 2:** Check for blown fuses or tripped breakers.
 - Yes: Replace fuses/reset breaker.
 - No: Move to next step.
- **Step 3:** Inspect windings for short circuits.
 - Yes: Repair or replace windings.
 - No: Move to next step.
- **Step 4:** Check for mechanical obstruction in the motor.
 - Yes: Remove obstruction.
 - No: Continue diagnosis based on motor symptoms.

5. Types of Faults in Machine Tools and General Causes**1. Mechanical Faults:**

- **Examples:** Misalignment, loose components, excessive vibration.
- **Causes:** Wear and tear, poor maintenance, incorrect installation.

2. Electrical Faults:

- **Examples:** Short circuits, faulty wiring, blown fuses.

- **Causes:** Power surges, insulation breakdown, improper wiring.

3. Hydraulic Faults:

- **Examples:** Leaks, low pressure, fluid contamination.
- **Causes:** Damaged seals, degraded fluid, clogged filters.

4. Pneumatic Faults:

- **Examples:** Air leaks, pressure drops, faulty valves.
- **Causes:** Loose connections, worn-out seals, contamination in air lines.

5. Thermal Faults:

- **Examples:** Overheating, inconsistent temperatures.
 - **Causes:** Blocked airways, poor ventilation, excessive load.
-

6. Conclusion

Summary:

Fault tracing is essential for maintaining efficient and safe industrial operations. By understanding the types of faults, the sequence of fault-finding activities, and how to use decision trees effectively, operators can minimize downtime, reduce costs, and ensure the longevity of equipment.

Key Takeaways:

- A systematic approach using decision trees improves diagnostic accuracy.
- Knowing the types of faults and their causes helps in quicker fault tracing.
- Consistent documentation and record-keeping facilitate future fault tracing.

Unit: V: PERIODIC AND PREVENTIVE MAINTENANCE

1. Define periodic maintenance

- Periodic maintenance is a process that ensures company assets remain in good condition throughout their useful life.
- It is based on the fixed maintenance schedule for assets like equipment, machinery, and vehicles.

2. What is the need for periodic maintenance

- Periodic maintenance service is important because it helps ensure that assets are long-lasting, stable, and reliable.
- Unlike other types of maintenance, periodic maintenance is nonselective.

3. What is degreasing.

- Degreasing is the process of removing grease, oil, and other types of contaminants from surfaces or parts.
- The process typically involves the use of a degreaser, which is a specialized cleaning agent designed to dissolve and remove these types of substances.

4. What is overhauling in machine.

- Overhauling of a machine is defined as a process of general maintenance performed on a machine or other industrial equipment.
- The goal of overhauling is to keep the system in serviceable condition.
- Regular checks can prevent all kinds of critical damage.

5. What are the benefits of overhauling

- Cost-effective
- Extended life length
- Increased performance
- Reduced labor costs

6. What are the common troubles in electric motor?

- Overheating
- Low resistance
- Electrical overload
- Vibration
- Contamination

7. What is meant by preventive maintenance?

- The care and servicing by personnel for the purpose of maintaining equipment and facilities in satisfactory operating condition by providing for systematic inspection, detection and correction of incipient failures either before they occur or before they develop into major defects.
- Maintenance, including tests, measurements, adjustments, and parts replacement, performed specifically to prevent faults from occurring.

8. Write any 4 daily checks of pumps

- Check pump exterior for any leaks
- Clean pump and nearby region to remove any debris
- Check for excessive pump vibration or unusual noises
- Check for foaming or oil discoloration

9. What is meant by repair cycle

- The repeated performance of all/some of the above mentioned activities in sequence between the successive overhauling is termed as “Repair cycle”.
- The repair cycle embraces the following activities.
 1. Inspection
 2. Repair
 3. Overhauling.

PART B & C**1. Explain in detail about concept, needs and repairing schemes of periodic maintenance.****Periodic Maintenance**

- Periodic maintenance is a process that ensures company assets remain in good condition throughout their useful life.
- It is based on the fixed maintenance schedule for assets like equipment, machinery, and vehicles.
- This type of maintenance heavily relies on the time interval given to the specific model of the asset.

Need

- Periodic maintenance service is important because it helps ensure that assets are long-lasting,
- stable, and reliable.
- Unlike other types of maintenance, periodic maintenance is nonselective
- It aims to inspect, clean, repair, and maintain every component of the equipment or machine which lessens the possibility of missing any factor that could become a problem.

Degreasing

- Degreasing is the process of removing grease, oil, and other types of contaminants from surfaces or parts.
- The process typically involves the use of a degreaser, which is a specialized cleaning agent designed to dissolve and remove these types of substances.
- Degreasers can be formulated with a variety of different chemicals, depending on the specific application and the type of contaminants that need to be removed.
- Common chemicals used in degreasers include:
 - Solvents like acetone or isopropyl alcohol
 - Alkaline cleaners like sodium hydroxide
 - Surfactants like detergents

2. Explain in detail about any one of the mechanical components.**Overhauling of Machine**

- Overhauling of a machine is defined as a process of general maintenance performed on a
- machine or other industrial equipment.
- The goal of overhauling is to keep the system in serviceable condition.
- Regular checks can prevent all kinds of critical damage.
- Machinery overhaul is usually performed by companies offering maintenance services.
- The frequency of overhauling can be agreed upon, routine maintenance is usually scheduled for once a year.
- A more frequent equipment check is recommended for older machines and especially larger machines involving complex mechanisms.

Benefits

- Cost-effective
- Extended life length

- Increased performance
- Reduced labor costs

Overhauling of a Machine in Stages

- Overhauling usually involves the following stages:
- **Inspection**
 1. First of all, the machine will be thoroughly inspected.
 2. Experienced maintenance crews perform an inspection on the overhauled machine under production conditions.
 3. It means, the machine's performance is monitored while the machine is in use.
 4. Such a procedure allows to allocate any issues and perform the troubleshooting more effectively.
- **Disassembly**
 1. After the initial inspection, the piece of equipment should be taken apart.
 2. Disassembly is crucial for further check and the next steps of the overhauling process, such as repair.
 3. A skilled maintenance worker is capable of putting the machine down efficiently, indicating which parts of the equipment need to be replaced or repaired.
- **Repair**
 1. Depending on the issue, the machine is either repaired or certain damaged parts are replaced.
 2. This step once again proves how effective overhauling is as opposed to replacing the whole piece of equipment at once.
 3. Replacement of parts might take longer than a simple repair, as the spare parts might need to be ordered from a manufacturer.
- **Reassembly**
 1. Following the successful replacement of spare parts, reassembly of the whole mechanism is performed.
 2. Being one of the final steps, the reassembly is crucial for the functioning of the equipment.
 3. Certain skill is surely needed to perform reassembly, so it's best handled by professionals.
- **Testing**
 1. The final step that concludes the overhauling process.
 2. Without testing it is naturally impossible to identify if the performed repair was effective.
 3. During testing the retrofit is either proclaimed successful or – less frequently – the process goes

back to the starting point (inspection).

4. Discuss about the overhauling of electric motor and common troubles involved in it with remedies.

Overhauling of Electric Motor

A standard electric motor overhaul includes an initial equipment inspection and diagnosis, bearings replacement, a test run and report.

A sample motor overhaul work scope includes –

- Collection from site.
- Inspect and record all relevant data from the nameplate.
- Carry out electrical and mechanical check tests to verify motor condition and any reported faults, where possible
- Dismantle motor
- Clean and inspect all component parts
- Datum checks, including bearing journals and seatings, shaft extensions, shaft extension run out, shaft seal fits, commutator diameter, and brush surface length
- Repair or replacement of defect components and parts
- Steam cleaning, stove drying and varnishing of stator and rotor windings as specified in IEEE1068
- Rotating parts dynamically balanced to ISO grade 2.5 or better
- Up to date motor plate fitted before dispatch
- Delivery service to customer site and recommissioning.

Common Troubles and Remedies of Electric Motor.

- Electric motors play a central role in machinery in every industry.
- A failure in this critical piece of machinery could be catastrophic to the business, with the potential for high costs and a long period of downtime.
- Understanding the common problems that can occur with electric motors can allow you to put measures in place to avoid failure and give your electric motor the best possible chance of achieving its maximum possible service life.

1. Low Resistance

- Low resistance is the most common cause of failure in electric motors. It is also often the most difficult to overcome.
- Under conditions such as overheating, corrosion or physical damage, degradation of the insulation of the internal windings of the motor may occur.
- This then causes insufficient isolation between the motor windings or conductors, leading to short circuits, leakages and eventually motor failure.
- Regularly inspect the insulation of the windings for signs of wear and replace before low

resistance leads to failure. If you are unsure, consult an expert.

2. Overheating

- Overheating is generally caused by either a high temperature in the operating environment or poor power quality.
- It is responsible for around 55% of insulating failures in electric motors. For every 10 degrees Celsius that the temperature of a motor rises, the insulation life is reduced by half.
- To avoid overheating, ensure that electric motors are kept as cool as possible.
- This can be done by maintaining as cool an operating environment as possible and regularly checking the temperature of the motor.

3. Electrical Overload

- Electrical overload is also commonly referred to as over current.
- It is caused by an excessive flow of current within the windings of the motor, which exceeds the design current that the motor is able to carry efficiently and safely.
- Over current is often the result of a low supply voltage, which results in the motor drawing in more current in an attempt to maintain torque.
- Electrical overload can also be caused by short-circuited conductors, or an excessive voltage supply.
- It is important to install effective over current protection which is able to detect overcurrent and interrupt supply to protect the motor.

4. Vibration

- Vibration can be extremely damaging to electric motors, frequently causing premature failure.
- It is often caused by the motor being positioned on an uneven or unstable surface.
- However, vibration can also be a result of an underlying issue with the motor, such as misalignment or corrosion. □
- Electric motors should be regularly inspected for vibration using a motor analyzing tool.
- Ensure that electric motors are positioned on a flat and stable surface.
- If vibration still occurs, check for signs of wear, as well as misalignment or loose bearings.
- Consider contacting a specialist if the source of vibration cannot be easily identified.

5. Contamination

- Electric motors frequently operate in environments where dust, dirt and chemicals are present, which may find their way inside the motor, leading to contamination and motor failure.
- These contaminants can dent bearing raceways and balls, leading to high levels of vibration and wear.

- They may also block the cooling fan, limiting the motor's ability to regulate its temperature and increasing the chances of overheating.
- As contamination is one of the leading causes of failure in electric motors, it is essential to prevent it from entering the motor. Luckily, this is relatively easy.
- Ensure that work areas, tools and fixtures are kept as clean as possible at all times to help eliminate the chances of contamination entering the motor.
- When laying out the workspace, try to position motors away from applications such as grinding machines which product large quantities of harmful contamination.
- Electric motors are the driving component of a vast range of applications across every industry, and regular inspection is essential to reduce the risk of premature failure.
- If you are in doubt about the condition of your motor, it is always advised to contact a specialist for further investigation.

5. Explain the concept of preventive maintenance and its advantages.

Preventive Maintenance

- The care and servicing by personnel for the purpose of maintaining equipment and facilities in satisfactory operating condition by providing for systematic inspection, detection and correction of incipient failures either before they occur or before they develop into major defects.
- Maintenance, including tests, measurements, adjustments, and parts replacement, performed specifically to prevent faults from occurring.
- The primary goal of maintenance is to avoid or mitigate the consequences of failure of equipment. This may be by preventing the failure before it actually occurs which Planned
- Maintenance and Condition Based Maintenance help to achieve.
- It is designed to preserve and restore equipment reliability by replacing worn components
- before they actually fail. Preventive maintenance activities include partial or complete overhauls at specified periods, oil changes, lubrication and so on.
- In addition, workers can record equipment deterioration so they know to replace or repair worn parts before they cause system failure. The ideal preventive maintenance program would prevent all equipment failure before it occurs.
- Preventive maintenance can be described as maintenance of equipment or systems before fault occurs. It can be divided into two subgroups:
- *Planned maintenance and*
- *Condition-based maintenance.*
- The main difference of subgroups is determination of maintenance time, or determination of moment when maintenance should be performed.

- While preventive maintenance is generally considered to be worthwhile, there are risks such as equipment failure or human error involved when performing preventive maintenance, just as in any maintenance operation.
- Preventive maintenance as scheduled overhaul or scheduled replacement provides two of the three proactive failure management policies available to the maintenance engineer.
- Common methods of determining what Preventive (or other) failure management policies should be applied are; OEM recommendations, requirements of codes and legislation within jurisdictions, what an "expert" thinks ought to be done, or the maintenance that's already done to similar equipment, and most important measured values and performance indications.
- Preventive maintenance is conducted to keep equipment working and/or extend the life of the equipment.
- Corrective maintenance, sometimes called "repair," is conducted to get equipment working again.

Examples in Some organization

- An individual bought an incandescent light bulb. The manufacturing company mentioned that the life span of the bulb is 3 years. Just before the 3 years, the individual decided to replace the bulb with a new one. This is called preventive maintenance.
- On the other hand, the individual has the opportunity to observe the bulb operation daily. After two years, the bulb starts flickering. The individual predicts at that time that the bulb is going to fail very soon and decides to change it for a new one. This is called predictive maintenance.
- The individual ignores the flickering bulb and only goes out to buy another replacement light bulb when the current one fails. This is called corrective maintenance

6. Explain the steps and procedure for periodic& preventive maintenance of machine tool, pump, air compressor and diesel generator.

Procedure for Periodic& Preventive Maintenance

1. Machine Tools

- 5 Preventive Maintenance Tips for Machine Tools
- Preventive maintenance of machine tools can be easily performed with a few basic tips, including the need to keep these tools:
- Dry by storing them in non-moisturized locations and safeguarding them from industrial rust and pollutants.
- Lubricated including the moving parts, internal, and fixed components without which, your machine tools are likely to break down due to wear and tear.
- Sharpened, particularly those tools that are used for cutting, sharpening, or slicing without

which, these machine tools are likely to have a shorter operating life due to constant wear and tear.

- Clean to get rid of industrial hazard materials and grime that can prevent your tools from working at optimum efficiency. In good shape by performing maintenance on smaller machine accessories such as the tool screws and bolts.
- With a digitized Front Line Worker Platform, manufacturers can optimize the preventive maintenance work of their machines and equipment. Let's look at how this can be accomplished.
- How Front Line Worker Platforms Improve Preventive Maintenance of Tools & Equipment
For effective preventive maintenance of tools and equipment in your manufacturing facility, front line operators must incorporate several actions as part of their daily tasks, including the following:
 - Regular inspection of machine tools for any defects or wear or tear
 - Complete cleaning of manufacturing assets including the factory premise and ventilation
 - Periodic lubrication of all moving parts in machinery to prevent damage
 - Repair or replacement of defective machine tools and equipment
 - A Connected Worker Platform improves preventive maintenance operations by empowering front line workers to:
 - Automate repetitive task assignment such as tool cleaning
 - Integrate the maintenance-related tips into their daily workflows and check off if they have been followed at the end of every working day.
 - Improve the quality of their maintenance work by streamlining best practices through an integrated platform and removing any redundant or inefficient steps from the process of cleaning their tools and equipment.
 - Implement daily inspection routines using a guided maintenance workflow as part of the PM guidelines.
 - Have procedures in place to make sure tools are properly collected at the end of jobs and that any issues without equipment is reported.

2. Pumps

- The strength and effectiveness of your pump preventive maintenance plan depending on how robust your checklist is. While you would want to include all possible checks in the plan, it is impractical and inadvisable for all routine checks.
- Therefore, the checklist is divided based on the frequency of checks making it more sustainable and effective:

Daily Check:

- Check pump exterior for any leaks

- Clean pump and nearby region to remove any debris
- Check for excessive pump vibration or unusual noises
- Check for foaming or oil discoloration
- Check bearing temperature for overheating
- Inspect all gaskets to ensure there are no oil leaks
- Inspect self flush pumps is applicable
- Clean bearing covers if needed
- Check pump cooling system

Monthly Check:

- Top up oil to bearing reservoirs if needed
- Clean oil bulbs and level windows
- Check the pump guards and replace them if needed
- Clear out dirt and debris from bearings and grease them
- If applicable, check that the hydraulic governors are working properly
- Check overall pump systems for leaks and clean the pumping system surroundings

3. Air Compressor

- Below is a checklist of things that you should check every time you maintain your air compressor.
- Typically, this relies on operating hours, so consult your owner's manual to see specific maintenance schedules.

1. Check the Oil

- This is very important for any engine (large or small).
- If your air compressor takes oil, make sure you check it before you start it up each time. If the oil is low, you could find yourself doing serious damage to the compressor. Whenever the oil is low, top it off.

2. Remove Moisture and Contaminants

- Filters, dryers, oil separators, and even the air tanks collect water, oil, and other contaminants from the air.
- The only way these components can operate efficiently is if you drain the waste from them.
- It should be done every time you've finished using the air compressor but needs to be inspected closely during maintenance.

3. Replace the Air Filter(s)—

- Frequently check all of your filters, and change them if there's heavy build-up. If you rarely use your air compressor, change the air filter every 3-6 months for optimal performance.

- If your air filter is dirty, your air compressor will work much harder to do its job. And even worse, you'll run the risk of contaminating the compressor.

4. Replace the Separator Element—

- The separator element prevents excessive use of oil.
- You may not want to pay for a replacement separator, but it's honestly cheaper than the rise in energy costs over time due to a drop in separator pressure.
- For every 2 PSI the separator pressure drops, your energy costs can increase by 1 percent.
- Periodically replacing your separator element will keep your compressor healthy and reduce how much you're spending on energy costs.

5. Clean the Fuel Tank—

- If you've got an oiled compressor, you'll need to clean the fuel tank during servicing to ensure it remains clear of gunk and contaminants to continue running effectively.

6. Test Safety Shutdown Features—

- At a minimum, this should be done each time a compressor is fully serviced.
- Run through your company's safety procedure and follow protocol to ensure the compressor will operate correctly according to the plan.

4. DIESEL GENERATOR

Lubrication Service

- The Engine oil must be checked while powering off the generator at regular intervals using a dipstick.
- Allow the oil in the upper portion of the engine to drain back into the crankcase and follow the engine manufacturer's recommendations for API oil classification and oil viscosity.
- Keep watching the oil level as near as possible to the full mark on the dipstick by adding the same quality and brand of oil.

Cooling System

- Check the coolant oil level during shutdown periods at the specified interval.
- Must be noted these points "remove the radiator cap after allowing the engine to cool, and, if necessary, add coolant until the level is about 3/4 in" And a More critical role in balancing diesel engines require a balanced coolant mixture of water, antifreeze, and coolant additives.
- Examine the exterior of the radiator for obstructions, and remove all dirt, grime or foreign material with a soft brush or cloth with caution to avoid damaging the fins.
- If available means, use the low-pressure compressed air or a stream of water in the opposite direction of normal airflow to clean the radiator.

Fuel System

- This is the important point when it comes to the maintenance of diesel generators.
- Diesel is subject to contamination and corrosion within a period of time is one year, and therefore regular generator set exercise is highly recommended to use up stored fuel before it degrades.
- The fuel filters should be drained at the designated intervals due to the water vapour that accumulates and condenses in the fuel tank.
- Better check regularly testing and fuel polishing may be required if the fuel is not used and replaced in three to six months.

Testing Batteries

- If the battery's charges reach the dead-end level is a common cause of standby power system failures.
- The battery must be kept fully charged and well-maintained at an all-time 40% to 100% to avoid regular testing and inspection to know the current status of the battery and avoid low battery levels.

Routine Engine Exercise

- Regular exercising keeps the engine parts lubricated and thwarts oxidation of electrical contacts, uses up the fuel before it deteriorates, and helps to provide diesel generator maintenance to reliable engine starting.
- Engine exercise is recommended to be executed 15 days once or 25 days once for a minimum of 30 min

Keep your Diesel Generator Clean

- Maintain your engine all-time nice and clean because it will be taken care of Oil drips and other issues.
- Check day-by-day hoses and belts that are in good condition or not.
- Frequent checks can keep better conditions and other nuisances from nesting in your equipment.

Exhaust system inspection

- In case of any leaks along the exhaust line which usually occur at the connection points, the welds, and the gaskets.
- Find the place of leakages and repair them immediately by a technician.

5. Explain the programming and scheduling concept of preventive maintenance.**Program and Schedule of Preventive Maintenance of****1. Mechanical Equipment.**

- Every PM program is developed to increase the equipment life and reduce unscheduled downtime.
- However, you should list all other objectives of developing a PM plan.
- That can include lowering costs, increasing savings, achieving minimal unplanned downtime, and minimal lost production opportunity time
- Other things can include reducing spare parts usage, reducing manufacturing interruptions, reducing labour costs, increase machine life spans, improving the quality of products, and maximizing manufacturing time per machine.
- **PM Schedule**
- Once you're done listing the proper PM procedures, you would upload them in your Computerized Maintenance Management System (CMMS).
- At this point, you have to develop a maintenance schedule, with each PM procedure scheduled to efficiently make use of the people and resources available.
- Preventive maintenance scheduling is done daily, weekly, monthly, quarterly, semi-annual, and annual PM events.
- Not all machines require daily or even weekly checks; however, they do need monthly, quarterly, and annual checks. Your preventive maintenance software keeps track of the maintenance checks for you.
- **Machine Lubrication Engineering in PM**
- Almost all of the manufacturer recommendations include the importance of lubricating rotating and reciprocating machine parts.
- Many maintenance professionals misunderstand the importance of lubrication in preventive maintenance; they consider the PM plan as a means of ensuring a lubrication program.□
- However, the reality is more complicated, and lubrication tasks require much more effort than is traditionally thought.
- You need to visually and physically inspect the machine and its components to write the appropriate PM procedures for the lubrication, for starters.
- **Preventive Maintenance Training**
- It's crucial for each team member to understand all the steps for preventive maintenance in machines.
- It's very much possible that individuals are either not utilizing the right tools for maintenance or doing it completely wrong.
- In the short-term, that won't matter much, but in the long term, it can lead to multiple equipment failures.
- **PM Program Management Plan**
- It's crucial to have a proper preventive maintenance management system in place.□

- The best way to go around that is to have a complete work order system in place that captures labor hours, materials, and task details.
- **Communication and Collaboration**
- The last step of any successful preventive maintenance plan for machines is to develop proper communication channels.
- Communicating the past success of PM programs is essential for reaffirming your PM team and developing two-way feedback channels.

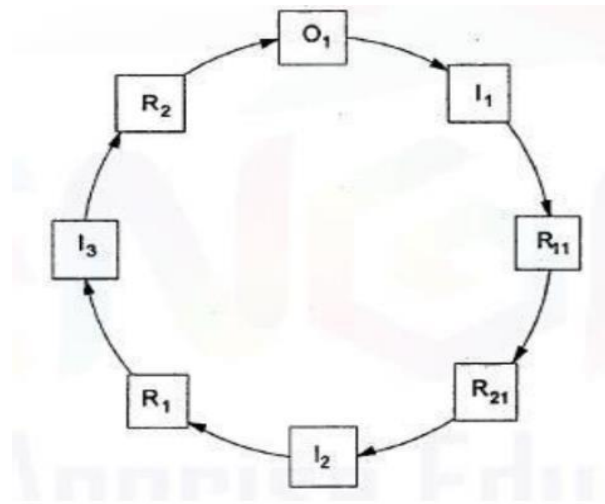
2. Electrical Equipment

- **Residential location checklist**
- Residential areas (also known as dwellings) usually have a smaller electric supply requirement. □
- This means fewer circuit panels and less wiring. □
- However, a preventive maintenance plan is just as important in a residence as it is in large facilities. □
- Check cable assemblies and other wiring methods in rooms, hallways, stairways, and attics. □
- Check cable installations and circuit boxes for general suitability.
- Examine boxes for warning lights and other irregularities.
- **Industrial and commercial establishment checklist**
- Check all power meters for regular function before shutting them down for maintenance. □
- Confirm that indicators, sensors, and warning lights are functioning correctly.
- Torque test and inspect bolted electrical connections to specified levels whenever needed.
- Visually inspect primary control points for signs of deterioration and overheating.
- Disassemble battery tripping packs and check for signs of general battery integrity and any signs of corrosion or heat damage. □
- Perform thermal imaging surveys using precise thermography equipment. □
- Check for signs of persistent overheating and single out specific areas for repair or part replacement. □

6. Explain about the concept of repair cycle.

Repair Cycle Concept and Importance

- The repeated performance of all/some of the above-mentioned activities in sequence between the successive overhauling is termed as “Repair cycle”. The repair cycle embraces the following activities. **1. Inspection 2. Repair 3. Overhauling.**



- It is clear from that first an inspection activity is scheduled followed by minor/major activities.
- Then an inspection takes place followed by a minor repair. Ageing a second inspection is followed by a major repair.
- Second inspection is followed by a major repair. Like this it goes and completes one repair cycle.
- The set of these activities between two consecutive overhauling is considered. More the
- complexity number more will be the activities involved and in turn more staffing requires to complete the repair cycle.